COHERENCE DEPTH AND AFFINE RECURRENCE IN A CUBICAL EXTENSION CALCULUS

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Abstract. We study framework extension over a fixed cubical object theory rather than over "cubical type theory" in the abstract. The theorem object is a cubical extension calculus built on a chosen CCHM-style cubical core with univalence and Kan composition, together with a discipline of globally acting opaque extensions whose public interfaces are counted only up to canonical telescope isomorphism. We call the least historical depth at which the normalized obligation sets cease to change, up to canonical equivalence, the *obligation depth* d_{obl}; separately, the *minimal-signature depth* d_μ records which primitive historical trace arities survive in canonical μ-minimal public form. Within this calculus we prove a payload-aware trace principle decomposing each sealed public interface into new core payload and resolved coherence trace. The full tagged normalized trace presentation retains primitive-versus-derived cost tags, while the presentation-invariant minimal opaque cost μ counts only the requires-primitive trace part. The cubical metatheory together with the bridge normalization results yields a depth-two theorem stack: exact stabilization of normalized obligation sets at depth 2; elimination of every primitive historical field of arity at least 3 from μ-minimal public signatures; recent-history factorization of every surviving binary primitive clause through the previous two normalized signatures; and a genuine binary obstruction showing that depth 1 does not suffice. The paper also isolates an abstract horn-computational interface strong enough to imply that exact-stabilization-plus-bridge pattern, and verifies that stronger package for the chosen CCHM-style cubical calculus. Abstracting that proof pattern yields a sparse footprint recurrence for minimal opaque cost; the fully coupled recurrence is the complete recent-history special case, not a descriptive law of arbitrary repository growth. In extensional systems with UIP the obligation depth collapses to 1. In the fixed fully coupled cubical extension calculus studied here the obligation depth is 2, while higher-coherence elimination shows $d_\mu \leq 2$, so

$$\mu_{n+1} = \mu_n + \mu_{n-1} + 2c$$

under constant payload size c; after the shift $U_n = \mu_n + 2c$, the sequence is Fibonacci. The accompanying Cubical Agda development formalizes the candidate-and-obligation calculus, the cubical exact upper- and lower-bound metatheory, the canonical-trace replacement step underlying μ-minimal elimination, and the abstract recurrence package; section 6 records the exact paper-to-Agda map and the remaining paper-level passage from raw surface declarations to the full tagged trace-presentation interface and its μ-primitive filter.

Key words and phrases: cubical type theory, extension calculus, coherence, univalence.

1. Introduction

1.1. **The metatheoretic question.** This paper concerns a simple but surprisingly rigid question.

> When a type-theoretic core calculus is extended by sealing a new structure against the existing interface, how does the cost of integration scale with the historical depth of the library?

The question is not about transparent user developments inside a fixed library. It is about *sealed* extensions: new type formers, modalities, or promoted interface packages whose public behavior becomes part of the global exported interface. More specifically, the paper studies the worst-case regime of *foundational core extensions*: globally acting additions whose exported payload must be specified across the currently active interface. Examples include extending a univalent universe hierarchy, where the decoding machinery must be updated against the library's active codes; adding a global modality presented by explicit modal dependent-type syntax, such as a Fitch-style dependent right adjoint or its cubical multimodal refinements [BCM+20, AKGB24], whose action must distribute across the exported type formers and transports already present; and heavily promoted generic interfaces whose action is intended to range over the whole library. In that setting, extension is constrained not only by local typing rules, but also by the obligation to interact coherently with the eliminators, transports, computation principles, and comparison data already exported by the library.

Orthogonal extensions are not the subject of this recurrence law. Adjoining a new datatype such as `Tree` to a library that already contains `List` and `Socket` need not introduce primitive `Tree`–`Socket` interaction data merely to preserve canonicity. Such sparse or local extensions are important in practice, but they do not exhibit the maximal coupling pattern that this paper isolates. Ordinary user-level library growth therefore forms a sparse dependency DAG in general; the recurrence theorem studies only the asymptotic foundational-core regime of monolithic, globally acting extensions.

The central observation is that these obligations are stratified by historical depth. Some are purely unary: the new layer must act on one previously exported generator. Others are genuinely binary: the new layer must preserve a previously exported compatibility witness between two generators. The paper therefore tracks two related depth notions: one for exact stabilization of obligation sets and one for what survives in normalized public signatures.

1.2. **The fixed theorem object.** This paper fixes one cubical object theory from the start. Let C be a CCHM-style cubical core calculus with univalence and Kan composition [CCHM18, CHM18]. Over C we study a language of globally acting extensions exported through opaque public signatures. A sealing step exposes only the normalized payload and trace fields of that public signature; the internal derivation that discharged the obligations is hidden behind opacity. The recurrence law proved below is therefore a theorem about a fixed cubical extension calculus built over C, not a blanket claim about every system that carries the label "cubical." Even for Cubical Agda, the claim is about this canonicalization and maximal-coupling discipline, not arbitrary library engineering. Cubical Agda enters in two roles only: as a mechanization environment and as a source of native cubical constructions used to witness the lower-bound phenomena.

1.3. **The core thesis.** The exact metatheoretic invariant is the *obligation depth* d_{obl}; the separate surface-syntactic invariant is the *minimal-signature depth* d_μ. The thesis of the paper is that, once obligation depth is isolated from minimal-signature elimination, the asymptotic law of integration becomes forced.

> In the fixed fully coupled cubical extension calculus of this paper, minimal opaque extension debt obeys a depth-two affine recurrence; under uniform payload size, the shifted sequence is Fibonacci.

Here "fully coupled" refers only to the foundational-core regime just described: admissible extensions are globally acting payloads whose semantics or promoted action is intended to range over the whole active interface, not orthogonal additions that leave most earlier exports untouched. Within that regime, the operative assumption is the scope of the advertised global action together with the active-basis contract of 2.22, not a theorem about arbitrary library growth. 2.26 identifies the primitive active *basis* interface of this fixed extension calculus: native canonicity-preserving totality and promoted operational exhaustiveness motivate the contract, while the formal density theorem proves the contract-to-density implication. The resulting recurrence is stated over the presentation-invariant minimal opaque cost μ, not over a chosen surface presentation. In extensional systems with UIP, the obligation depth collapses to 1, and under uniform payload size the recurrence is first-order affine. In the fixed cubical extension calculus studied here, the obligation depth is 2, and 4.13 gives the separate surface-syntactic bound $d_\mu \leq 2$; after the constant-payload shift the affine law becomes Fibonacci.

1.4. **Summary of contributions.** The paper develops one theorem package with three main components.

(1) We fix a cubical extension calculus over a chosen CCHM-style core, formalize its states, surface extensions, elaboration, sealed extensions, promoted interface packages, active interfaces, and integration latency, and distinguish orthogonal extensions from globally acting foundational core extensions. The payload and latency counts are defined on flattened telescope normal forms, so they are invariant under admissible interface refactorings such as rebundling records or splitting one interface into several shells. The bridge section then proves that elaboration, sealing, and counting normalization turn explicit typed raw declarations into full tagged canonical normalized public signatures, and then into a μ-primitive trace filter, so the abstract counts are theorems about extension syntax over the fixed cubical core. Within this framework we prove a basis-family coverage/minimality theorem for the fully coupled foundational-core regime, show that basis cardinality is independent of the chosen representatives of the transparent-generation classes, establish a payload-aware *Integration Trace Principle*, and define a presentation-invariant minimal opaque cost μ measuring the irreducible trace fields that survive in canonical primitive form.

(2) We define obligation depth d_{obl} for exact stabilization of normalized obligation sets and minimal-signature depth d_μ for elimination in μ-minimal normalized public signatures, and we prove the sharp cubical/extensional dichotomy for the exact invariant. In extensional type theory with UIP, the obligation depth is 1. In the fixed fully coupled cubical extension calculus, the obligation depth is 2, while 4.13 gives the separate bound $d_\mu \leq 2$. The depth-two obligation statement is supported by five complementary ingredients: a surface-to-horn normal-form theorem showing that normalized raw structural obligations

land in the horn-generated language; an exact stabilization theorem for normalized obligation sets showing $\mathcal{O}^{(k)}(X) \simeq \mathcal{O}^{(2)}(X)$ for $k \geq 2$; a separate surface-syntactic corollary, obtained by reading the cubical synthesis theorem through the bridge normalization results, showing that primitive historical fields of arity at least 3 are eliminated from μ-minimal public signatures; a recent-history factorization theorem showing that every surviving binary clause factors through the previous two normalized signatures; and lower bounds from an explicit swap-induced binary sealing obstruction, read categorically by adjunction coherence, together with a topological exact family supplied by clutching constructions for principal bundles over spheres. We also isolate the explicit horn-computational hypothesis package behind that exact-stabilization-plus-bridge argument; in this paper that stronger package is verified only for the chosen cubical calculus, while the weaker exact/window wrapper is what supports the separate strict/fibrant 2LTT-style example.

(3) We combine the trace principle with the obligation-depth theorem to obtain a sparse footprint recurrence for minimal opaque cost, with the fully coupled theorem as the complete recent-history envelope. Under constant payload size c, the depth-two cubical full-coupling case specializes to

$$\mu_{n+1} = \mu_n + \mu_{n-1} + 2c,$$

and the shifted sequence $U_n = \mu_n + 2c$ is Fibonacci. The associated structural inflation factor still converges to the Golden Ratio.

The Cubical Agda development described in section 6 formalizes the abstract candidate-and-obligation calculus, the cubical exact upper- and lower-bound metatheory, the contractible-factor decomposition of higher exact fibers, the recent-history factorization chain, the canonical-trace replacement interface for μ-minimal elimination, and the abstract recurrence package underlying 4.8, 4.10, 4.15, 4.16, 4.18, 5.8, 5.9, 4.19, 4.21, 4.23, 5.10, 5.3, 5.4, 5.5, and 5.7. The surface-to-horn image theorem 4.9, the sparse footprint reading of 5.2, and the remaining surface-syntactic readings of 2.26, 2.27, 3.12, 4.13, 4.17, and the μ-based recurrence are then obtained in the paper by combining those formal metatheorems with the bridge results of section 3. section 6 records the exact Agda theorem names and the trusted boundary.

Table 1 summarizes the dependency architecture that the rest of the paper keeps separate: exact stabilization of obligation sets, elimination from μ-minimal signatures, recent-history factorization, lower-bound binary obstructions, the affine recurrence itself, and the status of each layer in the mechanization.

1.5. **Related work.** The higher-dimensional background comes from the interpretation of intensional identity types as weak ω-groupoids, developed by Lumsdaine [Lum10] and van den Berg and Garner [vdBG11]. Several cubical calculi and implementations appear in the literature; the operational setting fixed in this paper is the CCHM-style cubical core of Cohen, Coquand, Huber, and Mörtberg [CCHM18], together with higher inductive types and Kan composition [CHM18]. Cubical Agda [VMA19] provides the mechanization environment and the native cubical programs used in the development, but the theorem target

Table 1: Dependency map of the depth-two theorem stack.

Claim	Logical role	Main inputs	Status
4.9	raw structural obligations land in the horn-generated normal form	bridge normalization, opacity, and factorization-complete trace export	paper-level bridge theorem
4.15, 4.16	exact $\mathcal{O}$-stabilization at depth 2 and trivialization of higher exact fibers	4.9, horn-reduction metatheory, and contractibility of based horn-extension fibers	formalized for the horn language; surface reading via bridge
4.13	elimination from μ-minimal normalized public signatures	4.9, 3.7, and canonical presentation	bridge corollary (replacement step formalized)
4.17	recent-history factorization of surviving binary primitive fields	factorization-complete trace export and the formal telescopic-subsumption chain	bridge reading of formal trace chain
4.19, 4.21	lower-bound binary obstruction	swap barrier, adjunction coherence, and clutching-family witness	fully formalized
5.2	sparse footprint law below the full envelope	active-basis coverage restricted to the advertised footprint and the trace-principle count	paper-level footprint reading
5.3	affine recurrence for μ	2.27, 3.10, and the chronological-window package	formalized abstractly, read through bridge
section 6	trusted-boundary audit	theorem-to-Agda correspondence for all rows above	mixed; see Table 4

of the paper is the extension calculus defined below, not the full Agda surface language. For modal examples we use explicit modal type-theoretic syntax rather than informal endofunctor language: dependent right adjoints give a canonical Fitch-style dependent presentation [BCM+20], and cubical multimodal type theory shows how such modalities interact with cubical structure [AKGB24]. For comparison with mixed weak/strict equality foundations, see the two-level presentations of Altenkirch, Capriotti, and Kraus and of Annenkov, Capriotti, Kraus, and Sattler [ACK16, ACKS23].

The coherence-theoretic side is informed by Mac Lane's coherence theorem [Lan63]. We also use Stasheff's analysis of higher associativity [Sta63] and the well-foundedness approach to coherence of Kraus and von Raumer [KvR19]. Our contribution is to turn coherence from a qualitative structural property into a quantitative invariant of extension debt.

The topological family used for the lower bound is the classical clutching description of principal bundles over spheres; for standard bundle-theoretic background we refer to Husemoller [Hus94]. The univalent and cubical perspective sits in the general landscape of homotopy type theory and univalent foundations [Uni13].

How to read this paper. The paper is organized in three layers that can be read at different speeds. *Layer 1* (Sections 1–2) sets up the fixed cubical extension calculus, its sealing discipline, and the presentation-invariant counts μ and κ; readers familiar with CCHM-style cubical type theory can skim the definitions and focus on the five-way taxonomy of 4.2, which governs every later proof. *Layer 2* (Sections 4–5) contains the main theorems: the depth dichotomy, the upper and lower bounds, the chronological window, and the affine recurrence. The key logical dependency is short: 4.9 puts raw structural obligations into horn normal form; 4.15 gives exact stabilization; 4.13 removes higher primitive trace fields from μ-minimal signatures; and 4.17 turns the remaining binary clauses into 4.18 (the two-layer window), which together with 2.27 drives the recurrence of 5.3; Table 1 summarizes the full dependency map. *Layer 3* (Sections 3 and 6) documents the bridge from surface syntax to canonical interface and the paper-to-Agda correspondence; these sections are important for evaluating the mechanization boundary but are not prerequisites for following the mathematical argument. Readers coming from categorical homotopy theory may find the lower-bound constructions (4.19, 4.21) the most natural entry point; readers coming from proof assistants may prefer to start with section 6 and read backwards.

2. A fixed cubical extension calculus

We fix throughout a cubical extension calculus built over a chosen CCHM-style core C with univalence and Kan composition. Its admissible extensions are globally acting payloads exported through opaque public signatures, and later layers may depend only on that public interface. The library therefore grows as a sequence

$$\mathcal{B}_0 \rightsquigarrow \mathcal{B}_1 \rightsquigarrow \mathcal{B}_2 \rightsquigarrow \cdots$$

of states, where each step adjoins one new sealed layer of the fixed extension calculus.

2.1. States, extensions, and promoted packages.

Definition 2.1 (State)**.** A *state* or *library* is a monotone context $\mathcal{B}$ closed under derivability. An evolution step

$$\mathcal{B}_n \rightsquigarrow \mathcal{B}_{n+1}$$

adjoins one sealed integration layer, denoted L_{n+1}.

Definition 2.2 (Typed raw surface extensions)**.** Over a state $\mathcal{B}_n$, a *typed raw surface extension declaration* is a finite package

$$e = (\Gamma_{\text{new}}, \Gamma_{\text{str}}, \Gamma_{\text{alg}}, \sigma, \mathcal{D}).$$

Here Γ_{new} is a finite payload telescope of new core symbols; Γ_{str} is a finite telescope of structural integration clauses; Γ_{alg} is a finite telescope of user-specified higher algebraic operations internal to the candidate; σ selects the public export presentation; and $\mathcal{D}$ is the typing and sealing derivation showing that the declared payload and structural clauses elaborate in the fixed cubical core and discharge the induced structural obligations.

Each clause of Γ_{str} is tagged by historical support and is generated by the grammar

$$c ::= \mathsf{act}(X, A) \mid \mathsf{cmp}(X, A, B) \mid \mathsf{horn}(\partial).$$

The constructors mean, respectively, unary action on a prior basis site, binary comparison against a prior comparison site, and a higher structural horn-extension clause determined by an already visible boundary ∂. There is no raw primitive constructor for arbitrary

closed higher structural fillers. If a user declares an arbitrary ternary or higher operation, it belongs to Γ_{alg} and is payload or payload-indexed structure, not structural integration trace. The admissible extensions studied in this paper are exactly those typed declarations whose intended action is global on the designated active interface footprint.

Remark 2.3 (No naked higher structural projections)**.** A surface presentation may name the missing remote face of a higher horn extension, but the typed structural clause is the whole horn-extension package

$$\sum_{\gamma} \mathsf{Fill}_{\partial}(\gamma),$$

or equivalently the based endpoint-plus-filler type displayed in 4.9. Thus an exported remote face of arity $k \geq 3$ is either retained only as a transparent abbreviation for the cubically synthesized face, or it is packaged with the hidden filler witnessing membership in the contractible horn-extension fiber. The calculus does not admit a naked opaque remote higher face as primitive structural trace.

Definition 2.4 (Elaboration, sealing, and normalized public signature)**.** Each admissible typed raw surface extension declaration e over $\mathcal{B}_n$ elaborates to a core object

$$\mathsf{Elab}(e)$$

of the fixed cubical core C: the payload telescope elaborates to new core declarations, the structural clauses elaborate to fields witnessing interaction with the active interface, the algebraic telescope elaborates to payload or payload-indexed internal structure, and the opacity component σ selects the public telescope exported after sealing. The *full canonical normalized public signature* of e is the tagged counting normal form

$$\widehat{N}(e) := \widehat{N}(\mathsf{Elab}(e)),$$

obtained by flattening the public telescope, choosing canonical representatives up to telescope isomorphism, and assigning primitive-versus-derived cost tags to trace fields. Transparent derived fields may remain in the full presentation as definitional aliases; they are simply not counted in the μ-primitive part below.

Definition 2.5 (Full trace presentation and μ-primitive part)**.** Two admissible typed raw surface extension declarations e and e' over the same state are *presentation-equivalent*, written

$$e \approx e',$$

when their elaborated sealed extensions determine the same public typing judgment up to definitional equality, canonical telescope isomorphism, and transparent internal normalization.

The full canonical normalized public signature decomposes as

$$\widehat{N}(e) = \widehat{N}_{\mathrm{pay}}(e) \uplus \widehat{N}_{\mathrm{tr}}(e),$$

where $\widehat{N}_{\mathrm{tr}}(e)$ is the full canonical trace presentation. Each field $\phi \in \widehat{N}_{\mathrm{tr}}(e)$ carries historical support, arity, and a primitive-cost tag

$$\mathrm{Supp}(\phi), \qquad \mathrm{arity}(\phi), \qquad \mathsf{cost}(\phi) \in \{\mathsf{derived}, \mathsf{requiresPrimitive}\}.$$

The μ-primitive trace part is

$$N^{\mu}_{\mathrm{tr}}(e) := \{\phi \in \widehat{N}_{\mathrm{tr}}(e) \mid \mathsf{cost}(\phi) = \mathsf{requiresPrimitive}\}.$$

The *minimal opaque cost* of e is

$$\mu(e) := |N^{\mu}_{\mathrm{tr}}(e)|.$$

Equivalently,

$$\mu(e) = \min\{\, |N^{\mu}_{\mathrm{tr}}(e')| \mid e' \approx e \,\},$$

and 3.6 shows that the displayed canonical representative attains this minimum.

Definition 2.6 (Candidate)**.** A *candidate* over a library state $\mathcal{B}$ is a pair

$$X = (X_{\mathrm{core}}, \mathcal{G}_{\mathrm{obl}}),$$

where X_{core} is the mathematical content proposed for addition and $\mathcal{G}_{\mathrm{obl}}$ is the family of normalized atomic coherence obligations induced when X is sealed against the active interface of $\mathcal{B}$. Every admissible typed raw surface extension declaration e determines such a candidate by elaboration.

Definition 2.7 (Counting normal form and atomic schemas)**.** Every public interface in this paper is counted only up to *canonical telescope isomorphism*: reassociation and flattening of iterated dependent sums, splitting or rebundling one single-constructor record into several nested records, and currying or uncurrying iterated function arguments when these preserve the same sealed typing judgment. Concretely, an exported interface is read as a dependent record, equivalently an iterated Σ-type whose field types may themselves begin with Π-telescopes. Its *counting normal form* is the fully flattened telescope obtained by reassociating the Σ-spine and expanding such record wrappers.

An *atomic schema* is one irreducible generating field of this counting normal form. A field is irreducible when its inhabitant is not uniformly synthesized from earlier fields by definitional equality, reflexivity, or the generic structural operations of the ambient calculus.

Remark 2.8 (Canonical normal forms and equivalences)**.** Throughout the paper, "canonical" means induced by the fixed counting normalizer and by the transparent telescope isomorphisms explicitly quotiented in 2.7. It does not mean that an arbitrary finite closure system has unique minimal generating subsets. When two presentations are canonically equivalent, the equivalence is the one obtained by normalizing both exported telescopes, matching the resulting normal-form fields, and transporting along the displayed definitional equalities, reindexings, and structural operations. In particular, basis choices below are choices of representatives for normal-form transparent-generation classes.

Definition 2.9 (Core payload size)**.** For a candidate X, let $P(X)$ be the finite set of atomic payload schemas appearing in the counting normal form of the public interface contributed directly by X_{core} before any coherence witnesses against earlier layers are added. Its cardinality

$$\kappa(X) := |P(X)|$$

is the *core payload size* of X.

Definition 2.10 (Sealed extension)**.** A *sealed extension* is a candidate whose obligations have been discharged and whose resolved data are exported only through an opaque public interface. Future extensions may use those exports, but may not inspect the internal derivations that produced them.

Definition 2.11 (Promoted interface package)**.** A *promoted interface package* over $\mathcal{B}$ is a finite package of constructively irreducible generators and interaction data that is exported as one new sealed layer. Such a package need not modify the reduction rules of the kernel.

What matters is that its public generators become part of the active exported interface for later sealing steps.

Definition 2.12 (Integration latency)**.** The *integration latency* of a candidate X over $\mathcal{B}$ is

$$\Delta(X \mid \mathcal{B}) := |\mathcal{G}_{\mathrm{obl}}|,$$

the number of irreducible atomic obligations that must be resolved before X can be exported as a sealed layer.

Remark 2.13 (Refactoring invariance of the counting notions)**.** Because κ, Δ, and later historical support are computed on counting normal forms, admissible refactorings such as bundling laws into one record, splitting one interface into several record shells, exporting curried rather than uncurried arguments, or pushing transparently derivable witnesses deeper into an internal term do not change the counts. Only a change in the irreducible primitive fields of the sealed typing judgment changes the metrics.

For a sealed layer L_k obtained from a candidate X_k, we write $S(L_k)$ for the set of atomic exported schemas in the counting normal form of its public interface and

$$\kappa_k := \kappa(X_k)$$

for its core payload size. The indexing is chosen so that Δ_k denotes the sealing cost of L_k.

Proposition 2.14 (Transparent user-level code lies outside the recurrence)**.** *Let U be a family of transparent definitions, lemmas, or record packages over $\mathcal{B}$ whose symbols are definitionally reducible using the existing operational semantics and are not promoted to a new opaque exported layer. Then U contributes zero integration latency and does not alter the active interface seen by later sealing steps.*

Proof. Such a development creates no new irreducible interaction site against the exported interface. All reductions remain inherited from the pre-existing library, and no new opaque generators are added to the global interface telescope. Hence U is ordinary growth inside one state, not a state transition of the form $\mathcal{B}_n \rightsquigarrow \mathcal{B}_{n+1}$. □

Definition 2.15 (Foundational core extension)**.** A candidate X over $\mathcal{B}_n$ is a *foundational core extension* if its payload is intended to act globally on the currently active interface: each atomic payload generator in $P(X)$ is operationally or semantically incomplete until its behavior has been specified against every generator in the designated coupling footprint inside the active exported interface. Typical examples are extensions of a universe-decoding apparatus, global modalities presented by explicit modal dependent-type syntax whose context action must distribute across existing type formers and transports [BCM+20, AKGB24], and heavily promoted generic interfaces whose exported action is library-wide.

Remark 2.16. Not every sealed extension is foundational in the sense of 2.15. A new datatype or API layer may be orthogonal to much of the existing library and therefore require only sparse interaction data. Such extensions may preserve canonicity without becoming globally coupled to earlier layers. The recurrence theorem of this paper is intentionally a worst-case statement about foundational core extensions, not a universality claim about all sealed library growth.

This scope restriction is essential for the basis-coverage theorem below. Ordinary user-level additions typically enlarge a sparse DAG of modules or datatypes and do not force primitive interaction with unrelated earlier exports. The maximal-coupling statement proved

here isolates the special foundational-core regime in which the new sealed layer is advertised as a *global* operation on the active interface and the active-basis contract requires that advertised action to be supplied on the primitive normal-form basis sites.

2.2. **Global action, basis coverage, and canonical minimality.** To speak about coupling to historical layers, it is convenient to parameterize the visible interface by depth.

Definition 2.17 (Historical interface)**.** For $k \geq 1$, the *historical interface of depth k* at stage n is

$$I_n^{(k)} := \biguplus_{j=0}^{k-1} S(L_{n-j}),$$

the tagged coproduct of the atomic schemas exported by the most recent k sealed layers in counting normal form.

Definition 2.18 (Basis families of the historical interface)**.** For each sealed layer L, let $\sim_{\mathrm{tr}}$ be the equivalence relation on the fields of $S(L)$ generated by definitional equality, uniform polymorphic instantiation, transport along already exported equalities, canonical telescope isomorphism, and the generic structural operations already quotiented in 2.7. A *basis site* of L is a $\sim_{\mathrm{tr}}$-class containing an irreducible normal-form field. A finite family

$$S_{\mathrm{bas}}(L) \subseteq S(L)$$

is called a *basis family* if it contains exactly one representative of each such basis site. Thus the word "basis" is used in this normal-form quotient sense, not as a claim that arbitrary finite closure systems have unique inclusion-minimal generating sets.

For $k \geq 1$, once a basis family has been chosen for each of the layers $L_n, \ldots, L_{n-k+1}$, the associated *historical basis of depth k* at stage n is

$$I_{n,\mathrm{bas}}^{(k)} := \biguplus_{j=0}^{k-1} S_{\mathrm{bas}}(L_{n-j}).$$

Its elements represent the *basis sites* on which a globally acting extension is primitively specified.

Theorem 2.19 (Basis existence and canonical bijection)**.** *For every sealed layer L in counting normal form, basis families in the sense of 2.18 exist. If $S_{\mathrm{bas}}(L)$ and $S'_{\mathrm{bas}}(L)$ are two such basis families, then counting normalization identifies a canonical bijection*

$$S_{\mathrm{bas}}(L) \cong S'_{\mathrm{bas}}(L).$$

Consequently the cardinality $|S_{\mathrm{bas}}(L)|$ depends only on the canonical normalized public signature of L, and for every depth k the historical basis cardinality $|I_{n,\mathrm{bas}}^{(k)}|$ is independent of the chosen basis families and stable under presentation-equivalent replacement of the participating layers.

Proof. Since $S(L)$ is finite, the finite quotient of its irreducible normal-form fields by $\sim_{\mathrm{tr}}$ is finite. Choosing one representative of each class gives a basis family. If two basis families choose different representatives, each representative lies in a unique transparent-generation class and the two choices are identified by the canonical normal-form equivalence of 2.8. Thus any two basis families are canonically bijective by matching representatives of the same class, and their cardinalities agree. Presentation-equivalent layers have canonically

isomorphic full tagged normalized signatures by 3.6, so the same class-by-class identification gives the historical invariance claim. □

Definition 2.20 (Basis coverage and canonical basis minimality)**.** An admissible foundational core candidate sealed against $I_n^{(k)}$ has *basis coverage* if, for any historical basis $I_{n,\mathrm{bas}}^{(k)}$ supplied by 2.18, every basis site receives at least one primitive interaction datum from the new extension. It is in *canonical basis-minimal form* if, again for any such historical basis, after elaboration and normalization each basis site contributes at most one requires-primitive public trace schema. By 2.19, these properties do not depend on which basis family is chosen.

Definition 2.21 (Fully coupled foundation)**.** A foundation is *fully coupled* if the admissible sealed extensions under study are foundational core extensions whose public signatures are advertised as acting on the whole active historical interface. Equivalently, the paper's admissible extensions in a fully coupled foundation are globally acting payloads whose intended coupling footprint is the whole active interface, not orthogonal additions with sparse dependency patterns. This definition fixes the scope of the action; the active-basis contract below is the additional admissibility condition saying that the advertised global action is actually supplied on each primitive active basis site.

Definition 2.22 (Fully coupled active-basis contract)**.** Let X be a foundational core candidate sealed against the historical interface $I_n^{(k)}$ in a fully coupled foundation. We say that X satisfies the *active-basis contract* if, for every basis site $b \in I_{n,\mathrm{bas}}^{(k)}$, the sealed public typing judgment of X contains a primitive interaction witness for b, unique up to the transparent-generation equivalence quotiented by counting normalization. For native core extensions, canonicity-preserving totality of the operational semantics is the intended source of this contract; for promoted interface packages, operational exhaustiveness of the exported global interface is the intended source. In this paper the contract is part of the fixed fully coupled extension calculus.

Theorem 2.23 (Active-basis contract entails basis coverage)**.** *Let $\mathcal{B}_n$ be a library state in a fully coupled foundation, let X be an admissible foundational core extension sealed against the historical interface $I_n^{(k)}$, and let $I_{n,\mathrm{bas}}^{(k)}$ be any historical basis for that interface. If X satisfies the active-basis contract of 2.22, then every basis site of $I_{n,\mathrm{bas}}^{(k)}$ receives at least one primitive interaction datum from X.*

Proof. This is exactly the existence component of 2.22. The uniqueness component will be used below only after passing to the canonical normal form and the μ-primitive filter. □

Remark 2.24 (Scope of the density assumption)**.** 2.23 is not a mechanized operational theorem saying that CCHM canonicity alone forces these interaction fields. It is the density consequence of the fully coupled active-basis contract. The operational stories are the intended justifications for adopting that contract in the fixed calculus: a native global computational extension that omits an active basis site leaves a class of terms whose behavior is not specified by the extended decoding or reduction machinery, while a promoted global interface package that omits such a site is not operationally exhaustive as an exported action on the active interface. Orthogonal extensions and sparse user-level library growth do not satisfy the hypotheses and are outside the recurrence theorem.

Proposition 2.25 (Canonical normalized presentations are basis-minimal)**.** *Let e be an admissible typed raw surface extension declaration over $\mathcal{B}_n$ whose candidate $X(e)$ is sealed against the historical interface $I_n^{(k)}$, and let $I_{n,\mathrm{bas}}^{(k)}$ be any historical basis for that interface. Then the μ-primitive trace part $N_{\mathrm{tr}}^{\mu}(e)$ contains at most one requires-primitive public trace schema for each basis site of $I_{n,\mathrm{bas}}^{(k)}$. Equivalently, redundant primitive clauses that differ only by transparent polymorphic compression, definitional equality, transport, or generic structural operations do not survive in the canonical normalized presentation.*

Proof. Suppose two primitive clauses of e correspond to the same basis site. By construction of 2.18, they induce the same public trace field up to definitional equality, transparent transport, uniform polymorphic instantiation, or the generic structural operations already quotiented by counting normalization. If both survived as distinct requires-primitive fields in $N_{\mathrm{tr}}^{\mu}(e)$, then the sealed public signature would contain presentation-equivalent duplicate primitive trace data. But 3.3 says that $\widehat{N}(e)$ retains the full visible public fields after normalization with primitive-cost tags, and 3.6 identifies a canonical normalized presentation for each presentation-equivalence class. Hence one of the two clauses is redundant or transparently derived and disappears from the μ-primitive part. Therefore at most one requires-primitive trace schema survives per basis site. □

Theorem 2.26 (Active-basis density law)**.** *Let $\mathcal{B}_n$ be a library state in a fully coupled foundation, let X be an admissible foundational core extension sealed against the historical interface $I_n^{(k)}$, and let $I_{n,\mathrm{bas}}^{(k)}$ be any historical basis for that interface. If X satisfies the active-basis contract of 2.22, then*

$$\Delta(X \mid \mathcal{B}_n) = |I_{n,\mathrm{bas}}^{(k)}|.$$

In particular, sealing cost counts irreducible basis clauses rather than the raw number of exported generators, and the right-hand side is independent of the chosen basis family by 2.19.

Proof. Coverage is exactly 2.23, using the active-basis contract as its hypothesis. Minimality is exactly 2.25 after passing from typed raw surface declarations to their full tagged canonical normalized public signatures and then filtering to the μ-primitive trace part. Since $\Delta(X \mid \mathcal{B}_n)$ counts irreducible primitive obligations in that filtered primitive presentation, every basis site contributes one and only one irreducible primitive trace field. Therefore

$$\Delta(X \mid \mathcal{B}_n) = |I_{n,\mathrm{bas}}^{(k)}|.$$

□

2.3. **Why the active-basis contract is not the recurrence.** The active-basis contract is a scope condition on an advertised global action. It says that, once an extension is presented as acting on the whole active interface, the action must be supplied on one representative of each primitive normal-form basis site. This is analogous to specifying a homomorphism on generators: the generator-level data are necessary to define the action, but they do not determine the depth of historical memory by themselves.

The recurrence appears only after three independent ingredients are combined:

(1) active-basis coverage, which gives one primitive clause per active basis site;
(2) trace export, which makes resolved obligations reappear as public trace fields;

Table 2: Examples and nonexamples for the active-basis contract.

Extension pattern	Scope of advertised action	Full contract?	Counting law
Universe code former	decoding and transport across active codes	yes	full recent-history envelope
Global modality	action on dependent types, transports, and type formers	yes	full or partial, depending on advertised scope
Promoted generic interface	public operator intended for the whole active interface	yes when globally advertised	classified by footprint
Transparent lemma package	no new opaque layer	no	zero integration latency
Orthogonal datatype	local datatype/API footprint	no full contract	sparse footprint law
Local plugin or API extension	selected dependency subset only	no full contract	sparse footprint law

(3) chronological window stabilization, which says only the recent layers remain primitive. Only the first item is supplied by 2.22 and 2.26. The second item is 2.27 and the bridge from raw declarations to canonical trace presentations. The third item is a cubical metatheorem: in this paper it is the depth-two window of 4.18, with the UIP one-layer collapse handled separately by 4.8. Thus the active-basis contract can hold in models whose window is one, two, three, or sparse; it is not a disguised Fibonacci assumption.

This distinction is important for interpretation. Ordinary library growth often lies in the sparse rows of Table 2. The theorem studied below concerns the globally advertised foundational-core endpoint, where the footprint is the whole active recent interface after cubical window factorization.

2.4. **The integration trace principle.** In the fully coupled foundational-core regime, the preceding theorem identifies sealing cost with the size of the active basis interface for extensions satisfying the active-basis contract. The next theorem explains how those resolved obligations reappear in the next layer's public export alongside the layer's own new mathematical payload.

Theorem 2.27 (Integration Trace Principle)**.** *Each sealed integration layer whose candidate satisfies the active-basis contract exports, in canonical basis form, a disjoint union of two kinds of public schema: the new payload contributed by its core content and the trace schemas corresponding to its resolved basis obligations. Equivalently, if L_k is the layer obtained by sealing a candidate of core payload size κ_k and coherence cost Δ_k, then there is a canonical decomposition*

$$S_{\mathrm{bas}}(L_k) \cong S_{\mathrm{core}}(L_k) \uplus S_{\mathrm{tr}}(L_k)$$

with

$$|S_{\mathrm{core}}(L_k)| = \kappa_k, \qquad |S_{\mathrm{tr}}(L_k)| = \Delta_k.$$

Hence

$$|S_{\mathrm{bas}}(L_k)| = \kappa_k + \Delta_k.$$

Proof. There are two logically distinct sources of basis schemas in the canonical normalized presentation.

First, the candidate's core content $X_{k,\mathrm{core}}$ contributes genuinely new public generators: a new type former, modality, promoted interface operator, or other primitive schema that did not previously exist in the library. By definition of core payload size, the number of these newly exported payload schemas is κ_k. These form $S_{\mathrm{core}}(L_k)$.

Second, by 2.26, sealing against the active interface under the active-basis contract contributes exactly one irreducible primitive interaction datum for each basis site of the active interface. Formally, a sealed layer is exported as a dependent record, equivalently an iterated Σ-type, whose fields consist of the new payload together with the coherence cells needed to type that payload against earlier exports. Opacity hides the inhabitants of those fields, but not the field-types occurring in the exported signature. Thus once such a datum is supplied and the layer is sealed, it no longer survives as an inspectable derivation visible to future extensions; it survives as a public trace field recording how the new payload acts on that older basis site. These schemas form $S_{\mathrm{tr}}(L_k)$, and there is one of them for each resolved basis obligation. Hence $|S_{\mathrm{tr}}(L_k)| = \Delta_k$.

Crucially, this decomposition is exact at the level of the sealed typing judgment. Replacing a primitive witness by a deeper internal derivation can change the hidden inhabitant of a trace field, but it cannot remove the corresponding field-type from the exported telescope without changing the public interface against which later layers are checked.

Because the layer is sealed, every basis schema of L_k is of one of these two kinds: either it is a newly introduced payload symbol, or it is the exported trace of a resolved obligation attaching that payload to prior interface data. The two kinds are disjoint, since payload symbols are primitive generators introduced by the new layer itself, whereas trace schemas are indexed by the prior basis interface against which the new layer was sealed. Therefore the canonical basis export decomposes as

$$S_{\mathrm{bas}}(L_k) \cong S_{\mathrm{core}}(L_k) \uplus S_{\mathrm{tr}}(L_k),$$

and the cardinality formula follows immediately. □

Remark 2.28. The trace principle counts kernel-level extensions and promoted interface packages uniformly. In both cases the public export has a payload part and a trace part. The only difference is the kind of primitive datum involved: for native extensions the payload contributes new reduction heads and the trace contributes defining clauses, whereas for promoted packages the payload contributes new exported operators and the trace contributes comparison or transport witnesses.

3. From cubical syntax to normalized extension signatures

The counting calculus of section 2 is intended to measure explicit extension declarations over the fixed cubical core C, not a freestanding payload calculus detached from syntax. This section spells out that bridge. Typed raw surface extensions elaborate to sealed core objects, sealing exports only their opaque public interfaces, and counting is performed on the tagged canonical normalized public signature of that export.

3.1. **The raw extension calculus as formal syntax.** The bridge theorem is deliberately stated for the paper's fixed raw extension calculus. It is not a parser theorem for arbitrary Cubical Agda files and it does not classify every higher operation a user might write. The raw syntax is the finite declaration package of 2.2: new payload symbols in Γ_{new}, structural integration clauses in Γ_{str}, user algebraic structure in Γ_{alg}, an export policy σ, and a typing/sealing derivation $\mathcal{D}$. The structural grammar has exactly the constructors

$$\mathsf{act}(X, A), \qquad \mathsf{cmp}(X, A, B), \qquad \mathsf{horn}(\partial),$$

where the first two are unary and binary interaction traces and the third is a higher horn-extension package over an already visible boundary. The syntax therefore separates *structural integration trace* from *payload or payload-indexed algebraic structure* before counting begins.

Proposition 3.1 (Structural role classification)**.** *Let e be an admissible typed raw surface extension declaration over $\mathcal{B}_n$. After elaboration and sealing, every normalized public field of $\widehat{N}(e)$ is classified in exactly one of the following roles:*

(1) *payload introduced by Γ_{new};*
(2) *payload-indexed algebraic structure introduced by Γ_{alg};*
(3) *unary or binary structural trace introduced by an* act- *or* cmp-*clause of Γ_{str};*
(4) *higher structural trace introduced by a $\mathsf{horn}(\partial)$-clause of Γ_{str}.*

In particular, an arbitrary ternary or higher user operation is not a structural trace merely because it has high arity. It is classified as payload or payload-indexed algebraic structure unless its typing derivation assigns it the structural role of a horn-extension clause.

Proof. The declaration package records the role of each clause before elaboration. Elaboration preserves that role: Γ_{new} contributes new core symbols, Γ_{alg} contributes internal algebraic fields, and Γ_{str} contributes only the three structural constructors listed above. Counting normalization then flattens, reassociates, curries or uncurries, transports along already exported equalities, and removes transparent aliases, but it never changes the source role of an irreducible field. Hence the displayed four-way classification is stable under the canonical telescope isomorphisms quotiented by 2.7. □

The theorem proved below is therefore nontrivial but scoped: structural integration clauses of the fixed calculus have a normalized horn image; higher algebraic payload operations are not part of that image theorem.

3.2. **Elaboration and public signatures.**

Theorem 3.2 (Elaboration soundness for sealed exports)**.** *Every admissible typed raw surface extension declaration e over a state $\mathcal{B}_n$ determines:*

(1) *an elaborated candidate*
$$X(e) := (X_{\mathrm{core}}(e), \mathcal{G}_{\mathrm{obl}}(e)),$$
(2) *a sealed cubical core object*
$$\mathsf{Seal}(\mathsf{Elab}(e))$$
exported as a sealed layer $L(e)$ through the opaque public interface selected by σ,
(3) *and a full canonical normalized public signature*
$$\widehat{N}(e) = \widehat{N}(\mathsf{Elab}(e))$$
obtained from that exported interface by counting normalization.

In particular, the candidate calculus of section 2 is the image of admissible typed raw surface extensions under elaboration and sealing.

Proof. By 2.2 and 2.4, the payload telescope Γ_{new} elaborates to new core declarations, Γ_{str} elaborates to structural trace clauses witnessing the required interaction with the active interface of $\mathcal{B}_n$, and Γ_{alg} elaborates to payload or payload-indexed internal structure. Admissibility means precisely that these clauses type-check and discharge the induced coherence obligations, so the result packages into a candidate $X(e)$ together with an opaque exported layer $L(e)$. Applying the tagged counting normal form of 2.5 to the resulting public telescope yields $\widehat{N}(e)$. Thus every admissible typed raw surface extension gives rise canonically to the candidate-and-sealed-layer data used in section 2. □

Theorem 3.3 (Adequacy of full tagged normalized public signatures)**.** *Let e be an admissible typed raw surface extension over $\mathcal{B}_n$ and let $L(e)$ be its sealed export. Then:*

(1) *every field of $\widehat{N}(e)$ comes from a field of the sealed public interface of $L(e)$; counting normalization introduces no new primitive public datum;*
(2) *every field of $\widehat{N}(e)$ is either a payload schema contributed by $\Gamma_{\mathrm{new}} \uplus \Gamma_{\mathrm{alg}}$ or a trace schema arising from a resolved structural obligation of $X(e)$;*
(3) *every payload or trace schema visible after sealing occurs as a field of the full canonical presentation $\widehat{N}(e)$, with derived trace fields tagged rather than counted as primitive;*
(4) *the synthetic quantities $P(X(e))$, $S(L(e))$, $\Delta(X(e) \mid \mathcal{B}_n)$, $\mu(e)$, and $\mathcal{O}^{(k)}(X(e))$ are exactly the corresponding payload, public-signature, abstract latency, minimal opaque cost, and obligation sets computed from $\widehat{N}(e)$, $\widehat{N}_{\mathrm{tr}}(e)$, and $N^{\mu}_{\mathrm{tr}}(e)$.*

Proof. Point (1) is immediate from 2.4 and 2.7: the normalization procedure only flattens the exported telescope, chooses canonical representatives, and tags trace fields, so it cannot create a new primitive field not already present in the sealed typing judgment. Point (2) is the surface-language form of the decomposition proved abstractly in 2.27: after elaboration and sealing, every visible field is either newly introduced payload, payload-indexed algebraic structure, or the public trace of a resolved structural interaction with prior interface data. Point (3) follows because counting normalization removes only presentation artifacts: reassociation of telescopes and transparent internal definitions. Derived trace fields are retained in $\widehat{N}_{\mathrm{tr}}(e)$ only as tagged canonical fields or transparent aliases, and are excluded from $N^{\mu}_{\mathrm{tr}}(e)$. Point (4) is then immediate from the definitions of payload size, public schema set, minimal opaque cost, integration latency, and induced obligation sets in section 2 and section 4: those quantities are computed exactly from the full tagged normalized public signature, its trace part, and the primitive-cost filter of 2.5. □

Proposition 3.4 (Necessity of irreducible public fields)**.** *Let e be an admissible typed raw surface extension over $\mathcal{B}_n$, and let ϕ be an irreducible payload field or a trace field of $\widehat{N}_{\mathrm{tr}}(e)$ tagged* requiresPrimitive*. If $e' \approx e$, then $\widehat{N}(e')$ contains a canonically corresponding field ϕ' with the same payload/trace status, historical support, arity, and primitive-cost tag. Equivalently, no presentation-equivalent declaration can delete a requires-primitive public schema from the sealed public signature.*

Proof. By presentation equivalence, e and e' determine the same sealed public typing judgment up to definitional equality, canonical telescope isomorphism, and transparent internal normalization. If the field ϕ were absent from $\widehat{N}(e')$, then either the public typing judgment of e' would genuinely omit that field, contradicting $e \approx e'$, or the field would have to be

uniformly reconstructible from the remaining exported data by the transparent operations quotiented by normalization. The latter case is exactly the derived tag, so it contradicts the hypothesis that ϕ is payload irreducible or trace-tagged requiresPrimitive. Hence ϕ survives in every presentation-equivalent normalized public signature, with its tags preserved. □

Theorem 3.5 (Raw-to-canonical normalized-signature bridge)**.** *Let e be an admissible typed raw surface extension declaration over $\mathcal{B}_n$ in the fixed fully coupled cubical extension calculus. Then elaboration and sealing determine:*

(1) *an elaborated candidate $X(e)$ in the abstract obligation language;*
(2) *a sealed cubical core object* $\mathsf{Seal}(\mathsf{Elab}(e))$*;*
(3) *a full canonical normalized public signature*
$$\widehat{N}(e) = \widehat{N}_{\mathrm{pay}}(e) \uplus \widehat{N}_{\mathrm{tr}}(e);$$
(4) *for every historical depth k, a canonical trace-signature object $S_k(e)$ together with a presentation $P_k(e)$ of $S_k(e)$ by the raw exported trace fields of e;*
(5) *a canonical isomorphism*
$$\widehat{N}^{(k)}_{\mathrm{tr}}(e) \cong \mathcal{O}^{(k)}(X(e))$$
preserving historical support and primitive-cost tags.

Moreover, the construction is sound and complete for presentation equivalence:
$$e \approx e' \quad \Longleftrightarrow \quad \widehat{N}(e) \cong \widehat{N}(e')$$
by a canonical telescope isomorphism preserving the payload/trace decomposition, historical support, arity, and primitive-cost tags.

Proof. The proof is by induction on the typed raw declaration and then by structural normalization of the exported telescope.

Payload declarations in Γ_{new} elaborate directly to new core declarations of the fixed cubical core. User-specified higher algebraic operations in Γ_{alg} elaborate as payload or payload-indexed structure internal to the candidate. Together these give $\widehat{N}_{\mathrm{pay}}(e)$.

Structural clauses in Γ_{str} elaborate to the trace part. A unary clause $\mathsf{act}(X, A)$ elaborates to the corresponding unary obligation of $X(e)$ against the exported basis site A. A binary clause $\mathsf{cmp}(X, A, B)$ elaborates to the corresponding binary obligation. A higher structural clause has the form $\mathsf{horn}(\partial)$ by the raw grammar. Its elaboration is therefore a cubical horn-extension package whose visible faces are determined by already exported lower-arity trace data and whose remaining datum is the remote endpoint together with the based filler. Thus every raw structural trace field has a canonical image in the obligation language of $X(e)$.

The telescope normalizer then performs only the allowed transparent steps: flattening of dependent records, reassociation of Σ-spines, currying or uncurrying of iterated arguments, definitional equality, transport along already exported equalities, and cubical structural operations. Hence it creates no new primitive public datum. It merely chooses canonical representatives for the payload fields and trace fields already present after sealing.

The primitive-cost tag is assigned as follows. A normalized trace field is tagged derived precisely when its inhabitant is uniformly synthesized from earlier normalized fields by the transparent structural operations of the ambient cubical calculus. Otherwise it is tagged requiresPrimitive. Since the raw structural grammar separates payload, unary action, binary comparison, and higher horn extension, this tagging is stable under canonical telescope isomorphism.

For each depth k, the depth-k trace fields therefore form a canonical trace-signature object $S_k(e)$, and the raw exported trace telescope selected by σ gives a presentation $P_k(e)$ of that canonical signature. The map from raw trace fields to the obligation language is an isomorphism after normalization because both sides are generated by the same typed structural clauses modulo exactly the same transparent telescope equations. It preserves support by construction, since support is read from the sealed layer indices mentioned by the normalized field, and it preserves primitive-cost tags by the definition of the tag assignment.

Finally, presentation equivalence is exactly equality of the sealed public typing judgment up to definitional equality, canonical telescope isomorphism, and transparent internal normalization. These are precisely the equations quotiented by $\widehat{N}(-)$. Hence two raw declarations are presentation-equivalent if and only if their canonical normalized public signatures are canonically isomorphic, with payload/trace decomposition, support, arity, and cost tags preserved. □

Proposition 3.6 (Canonical normalized presentations)**.** *If $e \approx e'$, then*

$$\widehat{N}(e) \cong \widehat{N}(e')$$

canonically, preserving payload fields, trace fields, historical support, arity, and primitive-cost tags. Consequently

$$N^{\mu}_{\mathrm{tr}}(e) \cong N^{\mu}_{\mathrm{tr}}(e')$$

and

$$\mu(e) = |N^{\mu}_{\mathrm{tr}}(e)| = |N^{\mu}_{\mathrm{tr}}(e')|.$$

Proof. By 3.5, presentation-equivalent raw declarations have canonically isomorphic full tagged normalized signatures, preserving the payload/trace decomposition, support, arity, and primitive-cost tags. Filtering both sides by the equation

$$\mathsf{cost}(\phi) = \mathsf{requiresPrimitive}$$

therefore gives a canonical isomorphism of μ-primitive trace parts. The cardinality of that filtered trace part is invariant under presentation equivalence. Since derived fields can be deleted by 3.8 below and fields tagged $\mathsf{requiresPrimitive}$ cannot be deleted without changing the canonical public typing judgment by 3.4, this invariant is exactly the minimum defining μ. □

Theorem 3.7 (Computational replacement on canonical trace presentations)**.** *Let X be a candidate over $\mathcal{B}_n$, let k be a history depth, and let S be the full canonical normalized trace presentation attached to the depth-k trace family of X, including primitive-versus-derived cost tags. Let P and Q be two presented trace telescopes whose counting normal forms are S, and let ϕ be a presented trace field of P. Suppose there is a lower-arity boundary subfamily ∂_ϕ and a cubically synthesized term*

$$t_\phi(\partial_\phi),$$

built only from hcomp, transp, hfill, *definitional equality, and the transparent structural operations already quotiented by counting normalization, such that after passage to S the field corresponding to ϕ is judgmentally reconstructed from lower-arity boundary data and is therefore tagged as transparently derived rather than primitive. Then:*

(1) *P and Q are presentation-equivalent representations of the same full canonical trace presentation;*
(2) *in the μ-primitive part of the canonical trace presentation represented by S, the field corresponding to ϕ is absent as an irreducible primitive trace field;*

(3) *every other irreducible primitive trace field of P corresponds canonically to one of Q.*

Proof. Because both P and Q normalize to the same full canonical trace presentation S, they are presentation-equivalent representations of one canonical public trace interface. By hypothesis, the normalized image of ϕ is reconstructed from lower-arity boundary data by the transparent cubical operations already quotiented by counting normalization, so its canonical primitive-cost tag is derived rather than requiresPrimitive. Therefore ϕ is excluded from the μ-primitive part of S, which retains only requires-primitive trace fields. Since primitive-cost tags are preserved under passage between presentations of the same canonical presentation, every other irreducible primitive field of P corresponds canonically to one of Q. Thus the replacement step changes only the presentation of the trace telescope, not the canonical primitive content measured by μ. □

Lemma 3.8 (Deletion of a transparently derived field)**.** *Let*

$$\Gamma, \phi : A(\Gamma)$$

be a normalized trace telescope, and suppose ϕ is tagged derived*. Then there is a canonical presentation equivalence*

$$\Gamma, \phi : A(\Gamma) \simeq_{\text{pres}} \Gamma.$$

If the surface presentation keeps a public name for ϕ, that name is retained only as a transparent abbreviation

$$\phi := t_\phi(\Gamma)$$

and does not contribute an atomic primitive field to N^μ_{tr}.

Proof. Since ϕ is tagged derived, there is a transparent synthesizer

$$t_\phi : \prod_{\gamma:\Gamma} A(\gamma)$$

built from earlier fields by the operations quotiented by canonical normalization. Projection gives

$$\pi : \sum_{\gamma:\Gamma} A(\gamma) \to \Gamma,$$

and the synthesizer gives a section

$$s : \Gamma \to \sum_{\gamma:\Gamma} A(\gamma), \qquad s(\gamma) = (\gamma, t_\phi(\gamma)).$$

The composite πs is judgmentally the identity. The other composite is identified with the identity by the derivation that classified ϕ as transparent. In the horn-extension case, this derivation is the contractibility of the based horn-extension fiber, with center computed by cubical composition, transport, and filling. Hence the two telescopes are presentation-equivalent.

A retained field name is therefore no longer an opaque exported projection; it is a transparent definition expanding to $t_\phi(\Gamma)$. Since μ counts only opaque primitive trace fields, such a name contributes zero to N^μ_{tr}. □

Remark 3.9 (Status of the replacement bridge)**.** 3.7 is formalized on the explicit canonical-trace-presentation interface: `Metatheory/ComputationalReplacement.agda` makes presentation equivalence, primitive-versus-derived tagging, and elimination from the μ-minimal part of a canonical trace presentation explicit. The paper-level part of section 3 is the

raw-to-canonical normalization passage from typed typed raw surface declarations to that canonical presentation interface, now isolated as 3.5; the repository does not mechanize a text parser for arbitrary user syntax.

Corollary 3.10 (Minimal opaque cost agrees with abstract coherence cost)**.** *For every admissible typed raw surface extension declaration e over $\mathcal{B}_n$ whose candidate $X(e)$ satisfies the active-basis contract of 2.22,*

$$\mu(e) = \Delta(X(e) \mid \mathcal{B}_n).$$

If e_k is a declaration whose sealing exports L_k and whose candidate satisfies the active-basis contract, if $\mu_k := \mu(e_k)$, and if $S_{\mathrm{bas}}(L_k)$ is any basis family of L_k, then

$$|S_{\mathrm{bas}}(L_k)| = \kappa_k + \mu_k.$$

Proof. By 3.3, the full trace presentation $\widehat{N}_{\mathrm{tr}}(e)$ records the public trace schemas arising from resolved structural obligations of $X(e)$, with transparently synthesized fields tagged as derived. By 3.6, $\mu(e)$ is exactly the cardinality of the requires-primitive subpresentation $N^{\mu}_{\mathrm{tr}}(e)$. By 2.26, using the active-basis contract hypothesis, there is one and only one requires-primitive trace schema for each basis site of the active interface, so that cardinality is precisely $\Delta(X(e) \mid \mathcal{B}_n)$. The final formula is then the trace-principle decomposition of 2.27, rewritten using $\mu_k = \Delta(X(e_k) \mid \mathcal{B}_{k-1})$; its right-hand side is well defined for any basis family by 2.19. □

Definition 3.11 (Factorization-complete public trace export)**.** Let L_k be a sealed layer with $k \geq 2$. We say that its public trace export is *factorization-complete* if, after choosing any admissible typed raw surface declaration whose sealing exports L_k and reading its canonical full tagged normalized trace presentation, the following data are available for the next remote-comparison step:

(1) the adjacent unary bridge from L_k to L_{k-1};
(2) the binary comparison trace witnessing the composite comparison from L_k to L_{k-2} through L_{k-1};
(3) the degenerate or structural faces needed to assemble the corresponding open 3-box, generated transparently by definitional equality, transport along already exported equalities, and the generic structural operations already quotiented by counting normalization.

Equivalently, the next horn-extension problem involving L_k can be posed using only exported public trace fields and transparent structural operations, without reopening the hidden derivation that discharged the original sealing obligations.

Theorem 3.12 (Sealed layers export factorization-complete trace data)**.** *Every sealed layer L_k with $k \geq 2$ in the fixed fully coupled cubical extension calculus has factorization-complete public trace export.*

Proof. Choose an admissible typed raw surface declaration e_k whose sealing exports L_k. By 2.27, the public export of L_k decomposes into new payload and trace, and by 3.3 every irreducible trace schema visible after sealing appears in the canonical full trace presentation $\widehat{N}_{\mathrm{tr}}(e_k)$. Because L_k was sealed as a globally acting layer against the active interface, its resolved trace obligations include the adjacent action on L_{k-1} and the binary comparison against the already exported comparison between L_{k-1} and L_{k-2}. Hence the corresponding unary bridge and binary comparison trace occur as public fields of $\widehat{N}_{\mathrm{tr}}(e_k)$.

The remaining faces needed for the next open 3-box are not extra primitive obligations. They are degenerate composites, reindexings, and transports along already exported equalities, so they are generated transparently by the structural operations already quotiented in counting normalization. By 3.4 and 3.6, passing to a presentation-equivalent declaration cannot delete the irreducible unary or binary trace fields just identified; it can only rewrite them into canonically isomorphic normalized form. Therefore the factorization data are exported stably at the level of full tagged canonical normalized public signatures and can be reused without reopening the hidden sealing derivation. This is exactly factorization completeness. □

Remark 3.13 (Syntactic scope of the bridge)**.** The results of 3.2, 3.3, 3.5, 3.4, 3.6, 3.8, and 3.7 are syntactic bridge theorems: they speak only about elaboration, sealing, opacity, counting normalization, and the presentation-equivalence relation of 2.5. They justify that $\widehat{N}(e)$, $N^{\mu}_{\mathrm{tr}}(e)$, and $\mu(e)$ are invariants of explicit typed extension declarations over the fixed cubical core C.

By contrast, 2.26, 2.27, and the later depth and recurrence theorems are metatheorems of the abstract candidate-and-obligation calculus. Their application to surface syntax passes through the adequacy theorem and raw-to-canonical bridge above: once an admissible declaration has been elaborated and normalized, those later counts are read off from $\widehat{N}(e)$ and the requires-primitive filter $N^{\mu}_{\mathrm{tr}}(e)$. Throughout the rest of the paper we keep the definitions of $\widehat{N}(e)$, $N^{\mu}_{\mathrm{tr}}(e)$, and $\mu(e)$ from 2.4 and 2.5 unchanged; all later minimality statements are therefore relative to this fixed presentation-equivalence relation and cost-tag filter.

The remaining bridge obligation is not a parser theorem for all of Cubical Agda. It is an image theorem for the fixed extension calculus defined in section 2. The structural part of that calculus has no primitive constructor for arbitrary closed higher fillers. Its higher integration clauses are generated only by extending an already visible cubical boundary by one remote face and a filler. Consequently the normalized image of raw structural syntax is exactly the horn-generated obligation language isolated in 4.9. User-specified higher algebraic operations are classified as payload; they are not structural integration obligations and are therefore outside the stabilization claim.

4. Obligation Depth, Minimal-Signature Depth, and the Dichotomy

4.1. Induced obligations and historical arity.

Definition 4.1 (Induced obligations)**.** Let $\mathcal{O}^{(k)}(X)$ be the set of normalized atomic obligation types induced when candidate X is sealed against a historical interface of depth k. Each element is read in the counting normal form of 2.7: we quotient by the same canonical telescope isomorphisms and retain only irreducible generating fields not uniformly synthesized by the ambient calculus. Here normalization retains exactly the developer-visible data not already exported by prior layers. In particular, when a potential higher-dimensional coherence problem appears with one face left free, the corresponding atomic obligation is represented by the based horn-extension total type of the missing face together with a filler from the fixed lower cap, rather than by the type of fillers for a completely fixed boundary. Since the interface grows cumulatively, there are inclusions

$$\mathcal{O}^{(1)}(X) \subseteq \mathcal{O}^{(2)}(X) \subseteq \mathcal{O}^{(3)}(X) \subseteq \cdots .$$

Remark 4.2. The paper uses the following five-way taxonomy.

Status	Meaning
Present obligation type	A normalized obligation type genuinely appears in some $\mathcal{O}^{(k)}(X)$ and therefore belongs to the exact metatheoretic stabilization problem.
Contractible obligation type	The obligation type is still present in $\mathcal{O}^{(k)}(X)$, but its fiber is canonically contractible; this is not yet a statement about minimal public syntax.
Transparently derived field	A field is part of the sealed typing judgment but is uniformly reconstructible from other exported data by definitional equality, transport, or the generic structural operations already quotiented by counting normalization.
Eliminated from a μ-minimal signature	A primitive field may occur in some presentation, but every canonical μ-minimal normalized public signature removes it after passage through presentation equivalence and the bridge theorem.
Absent from the obligation set	No normalized obligation type of that kind occurs at all.

Obligation depth d_{obl} is a statement about exact normalized obligation types. Minimal-signature depth d_μ asks a different question, namely which primitive trace fields survive in canonical normalized public form. A bound on d_μ therefore does not by itself produce a chronological window. Accordingly, "contractible", "transparently derived", "eliminated from a minimal presentation", and "absent" are never used interchangeably in this paper. The schematic separation in 4.11 below is the motivating toy case: a depth-three horn factor remains present in the exact obligation set, is contractible as a type, and contributes no irreducible primitive field to μ.

Definition 4.3 (Historical support and arity)**.** For a normalized obligation o, let $\mathrm{Supp}(o)$ be the set of sealed layers whose atomic exported schemas occur irreducibly in the counting normal form of the typing derivation of o after each prior layer has been collapsed to its opaque interface. Its *historical arity* is

$$a(o) := |\mathrm{Supp}(o)|.$$

Remark 4.4. Historical arity counts independent prior layers, not textual age. A direct lookup to an old exported theorem is still unary, because it lands in one opaque interface. Multi-layer debt arises only when the new layer must cohere simultaneously with several independent prior exports.

Lemma 4.5 (Arity-to-dimension correspondence)**.** *An irreducible obligation of historical arity k requires a k-dimensional coherence cell. Arity 1 produces substitution or transport data, arity 2 produces a naturality square or homotopy, and arity $k \geq 3$ would require a genuine higher filler.*

Proof. With one historical input, the new layer must supply a comparison morphism between its behavior and one prior export, so the obligation is one-dimensional. With two independent historical inputs, the witness compares two unary composites and therefore has the form of a square or homotopy. More generally, relating k independently supplied historical interfaces requires a k-dimensional filler between the corresponding composites. □

4.2. Definition of obligation depth and minimal-signature depth.

Definition 4.6 (Obligation depth and minimal-signature depth)**.** A foundation, equivalently a fixed extension calculus together with its sealing discipline, has *obligation depth* d_{obl} if d_{obl} is the least integer such that for every candidate X and all $k \geq d_{\mathrm{obl}}$ there is a canonical equivalence

$$\mathcal{O}^{(k)}(X) \simeq \mathcal{O}^{(d_{\mathrm{obl}})}(X).$$

It has *minimal-signature depth* d_μ if d_μ is the least integer such that every admissible typed raw surface extension declaration e is presentation-equivalent to a declaration e' whose μ-primitive trace part $N^{\mu}_{\mathrm{tr}}(e')$ contains no historical trace field of arity greater than d_μ.

When the paper says simply "coherence depth," it means obligation depth, not yet a chronological-window theorem.

The point of 4.6 is that beyond depth d_{obl}, the normalized obligation types themselves stop changing up to canonical equivalence. This is stronger than saying merely that deeper history contributes no new minimal opaque cost or that d_μ is bounded. Once that exact stabilization has been upgraded to a genuine chronological window, the size of the primitive active basis interface controls the recurrence law for integration debt.

Remark 4.7. 4.6 records two depth notions. The obligation-depth bound d_{obl} is a stabilization statement about *obligation sets*; the minimal-signature bound d_μ is a statement about the μ-primitive part of full tagged canonical normalized public signatures. Neither should be read as saying that a bound on coherence dimension automatically identifies the chronological layers that must still be inspected primitively. In particular, an arity-two obligation could in principle couple a new layer to a remote historical layer. To extract a genuine time window from a dimensional bound one needs an additional factorization theorem showing that such remote binary comparisons are already generated by recently exported traces. In the cubical univalent case, this passage from arity to chronology is supplied by 4.17.

4.3. Theorem A: extensional systems collapse to $d_{\mathrm{obl}} = 1$.

Theorem 4.8. *In Martin-Löf type theory with UIP, the obligation depth is* 1*.*

Proof. Under UIP, any two proofs of identity between fixed endpoints are equal. Hence every identity type is a set, and every higher identity type is a proposition; once inhabited, it is contractible. Consider any normalized coherence obligation of dimension at least 2. By 4.5, it corresponds to a higher identity witness. But in the presence of UIP such a witness is uniquely determined by its boundary and therefore carries no independent data. Thus every obligation of arity at least 2 collapses to data already fixed by its unary boundary. Equivalently,

$$\mathcal{O}^{(k)}(X) \simeq \mathcal{O}^{(1)}(X) \qquad (k \geq 1).$$

Since depth 0 would mean the absence of any interaction with the current interface, which is impossible for a nontrivial sealing step, the least stabilizing obligation depth is 1. □

4.4. **Theorem B: the fixed cubical calculus has $d_{\mathrm{obl}} = 2$.** From now on we specialize the paper's fixed theorem object to the fully coupled cubical extension calculus built over the chosen CCHM-style core [CCHM18, CHM18], with univalence and Kan composition available in the ambient operational semantics. The next ingredients are deliberately complementary: an exact stabilization theorem for normalized obligation sets gives the bound $d_{\mathrm{obl}} \leq 2$; 4.13 gives the separate syntactic bound $d_\mu \leq 2$ on μ-minimal public presentations; recent-history factorization turns the exact depth bound into a genuine two-step time window; and the explicit swap obstruction together with the clutching family give lower-bound witnesses. Together they determine exact obligation depth 2.

Univalence and cubical computation play different roles in that argument. Univalence supplies the lower-bound source of genuine binary coherence, witnessed below by the swap obstruction and the clutching family. It does not by itself remove primitive arity-3 trace fields or produce exact stabilization of normalized obligation sets. Those exact-stabilization and elimination statements use the separate cubical horn-computation package together with recent-history factorization.

One semantic clarification is important before the cubical argument begins. In univalent foundations, types behave as weak ω-groupoids, so the geometric space of fillers for a fixed higher boundary is not generally contractible. The paper therefore does not model a depth-three obligation as the space of fillers for a closed 3-sphere. After the normalization of 4.1, the marginal higher-dimensional datum appears instead as a horn-extension problem: an open box with one remote face still free.

It is equally important to delimit the class of obligations under discussion. The paper does not claim that arbitrary internal higher-algebraic structure carried by a candidate disappears from cubical type theory. What is measured here is only *structural integration coherence*: the obligations induced by sealing a globally acting foundational candidate against prior opaque layers. Those obligations arise from the chronological dependency graph of exported payload and trace data, not from arbitrary user-specified higher operations internal to the candidate.

Concretely, once the visible unary and binary traces have been fixed, the depth-k marginal fiber is not a closed filler space. It is the total type which chooses the missing remote face and a filler from the fixed lower cap:

$$H_k(\partial) := \sum_{\gamma : A_\partial[\varphi_\partial \mapsto u_\partial(i1)]} \mathrm{PathP}(\lambda i.\, A_\partial[\varphi_\partial \mapsto u_\partial(i)], u_{0,\partial}, \gamma)\,.$$

Here ∂ is the already exported boundary, A_∂ is the cubical type determined by that boundary after any necessary transp alignment, φ_∂ is the visible face formula, u_∂ is the partial element on the visible faces, and $u_{0,\partial}$ is the lower cap. This based endpoint-plus-filler type is a horn-extension singleton; the paper never uses the false claim that all fillers for an arbitrary closed higher boundary are unique. In cubical type theory, the operational semantics provide generic Kan operations such as hcomp, comp, and transp that solve this based extension problem uniformly. The sharp question for this paper is therefore syntactic: does a field of that shape survive as an irreducible trace field in a μ-minimal normalized public signature, or can it be replaced definitionally by a term synthesized from lower-arity boundary data?

Theorem 4.9 (Surface-to-horn normal form for structural integration)**.** *Let e be an admissible typed raw surface extension declaration over $\mathcal{B}_n$ in the fixed fully coupled cubical extension calculus, and let $X = X(e)$ be its elaborated candidate. For every $k \geq 3$, the normalized raw structural obligations induced by sealing X against $L_n, \ldots, L_{n-k+1}$ are canonically equivalent*

to

$$\sum_{\partial:\mathcal{O}^{(k-1)}(X)} H_k(\partial).$$

For each boundary $\partial : \mathcal{O}^{(k-1)}(X)$, the fiber $H_k(\partial)$ is the based cubical horn-extension type

$$H_k(\partial) := \sum_{\gamma:A_\partial[\varphi_\partial \mapsto u_\partial(i1)]} \mathrm{PathP}(\lambda i.\, A_\partial[\varphi_\partial \mapsto u_\partial(i)], u_{0,\partial}, \gamma)\,,$$

where A_∂ is a cubical type, made homogeneous by transport when needed, φ_∂ is a face formula, the partial element

$$u_\partial : (i : I) \to \mathrm{Partial}(\varphi_\partial, A_\partial),$$

and the lower cap

$$u_{0,\partial} : A_\partial[\varphi_\partial \mapsto u_\partial(i0)]$$

are determined functorially by the already exported unary and binary trace data of the intervening sealed layers. The equivalence preserves historical support, primitive-versus-derived cost tags, and canonical telescope isomorphism. Under this equivalence, the depth-k marginal data are exactly the endpoint-plus- filler data of the displayed horn extension.

Proof. We argue by induction on the raw derivation of structural integration obligations after elaboration and sealing.

First, opacity restricts what later layers can inspect. A sealed layer contributes only its normalized public payload fields and normalized public trace fields. No later sealing step can use the hidden derivation that produced those fields. Hence every raw structural obligation is built from the newly declared payload of X, exported unary action traces, exported binary comparison traces, and the transparent structural operations of the ambient cubical calculus: definitional equality, reindexing, transport, homogeneous composition, and filling. The arity-one and arity-two cases are exactly the primitive unary and binary integration clauses retained in $\mathcal{O}^{(1)}(X)$ and $\mathcal{O}^{(2)}(X)$.

Now suppose $k \geq 3$ and the raw obligation has support $X, L_n, \ldots, L_{n-k+1}$. Passing from depth $k-1$ to depth k exposes only the new remote layer L_{n-k+1}. By the induction hypothesis, all obligations involving the intervening layers have already been normalized to exported unary and binary trace data. Because those layers are sealed, all faces of the resulting comparison cube except the direct remote face are therefore visible in the canonical public interface. The remaining raw datum is precisely a missing remote face γ together with a filler against the visible boundary.

That visible boundary determines a cubical type A_∂, after transp if needed, a face formula φ_∂, a partial element u_∂ on the visible faces, and a lower cap $u_{0,\partial}$. Therefore the normalized marginal obligation is the displayed based horn-extension type $H_k(\partial)$.

Conversely, a point of $H_k(\partial)$ supplies exactly the data needed for a raw structural integration clause: insert γ as the remote comparison face and use the dependent path component as the higher compatibility filler. The two constructions are inverse up to the canonical telescope isomorphisms and transparent cubical normalizations already quotiented by the counting normal form. Support preservation is immediate: the boundary part has support $X, L_n, \ldots, L_{n-k+2}$, and the horn-extension step adds exactly L_{n-k+1}. Primitive-cost preservation follows from the same normal-form grammar: arity-one and arity-two action/comparison fields are primitive, while the displayed horn-extension field is reconstructed from lower-arity boundary data using ambient cubical operations and is therefore tagged as derived. □

Lemma 4.10 (Syntactic horn reduction for structural integration coherence)**.** *Let X be a foundational core candidate sealed at stage $n+1$ in the fixed fully coupled cubical extension calculus. For every $k \geq 3$ and every boundary $\partial : \mathcal{O}^{(k-1)}(X)$, the additional data exposed by adjoining the remote layer L_{n-k+1} are canonically represented by the based horn-extension fiber $H_k(\partial)$ of 4.9. Here the endpoint γ is the new remote comparison face between X and L_{n-k+1}, while every other face of the box is already exported by the unary and binary traces of the intervening sealed layers.*

Proof. For a candidate presented by an admissible declaration, this is exactly the surface-to-horn normal form of 4.9. For an abstract representative of the same candidate class, the statement is read in the counting normal form of 4.1; canonical telescope isomorphism and 3.6 make the choice of surface representative irrelevant. □

A concrete cubical sketch. The abstract distinction between payload and trace can be made visible in a small cubical example. Suppose an older sealed layer exports a small Tarski universe with function codes and their decoding equation. A new cohesive layer contributes a genuinely new modal payload `Flat` together with its unit `eta`. Everything else needed to make that modality act on the older universe is integration trace: a lifted action on old codes, a comparison between decoding before or after lifting, and the distribution law over old function codes. In the snippet below, the witness `remote-face` is the point of the example. It is not additional payload. It is exactly the missing comparison face of the open 3-box determined by the old decoding law together with the new unary and binary trace data, and cubical composition computes it uniformly.

```
-- Schematic Cubical Agda: old universe + new payload + induced trace

record OldU : Set1 where
  field
    U      : Set
    El     : U -> Set
    Arr    : U -> U -> U
    El-Arr : (A B : U) ->
      Path Set (El (Arr A B)) (El A -> El B)

record Payload : Set1 where
  field
    Flat : Set -> Set
    eta  : {A : Set} -> A -> Flat A

record Trace (L : OldU) (P : Payload) : Set1 where
  open OldU L
  open Payload P
  field
    FlatU         : U -> U
    flat-El       : (A : U) ->
      Path Set (El (FlatU A)) (Flat (El A))
    flat-code-Arr : (A B : U) ->
      Path U (FlatU (Arr A B)) (Arr (FlatU A) (FlatU B))
```

```
    flat-Arr      : (A B : U) ->
      Path Set (Flat (El A -> El B))
               (Flat (El A) -> Flat (El B))

module _ (L : OldU) (P : Payload) (T : Trace L P) where
  open OldU L
  open Payload P
  open Trace T

  -- left route:
  --   El (FlatU (Arr A B))
  --     -> Flat (El (Arr A B))
  --     -> Flat (El A -> El B)
  --     -> Flat (El A) -> Flat (El B)

  -- right route:
  --   El (FlatU (Arr A B))
  --     -> El (Arr (FlatU A) (FlatU B))
  --     -> El (FlatU A) -> El (FlatU B)
  --     -> Flat (El A) -> Flat (El B)

  postulate
    left-route  :
      (A B : U) ->
      Path Set (El (FlatU (Arr A B))) (Flat (El A) -> Flat (El B))
    right-route :
      (A B : U) ->
      Path Set (El (FlatU (Arr A B))) (Flat (El A) -> Flat (El B))

  -- The open 3-box says these two boundary composites agree.
  -- This face is transparently derived trace, not new payload.
  remote-face :
    (A B : U) ->
    Path (Path Set (El (FlatU (Arr A B)))
                   (Flat (El A) -> Flat (El B)))
         (left-route A B) (right-route A B)
  remote-face A B = hcomp ...
```

Read top-to-bottom, `Payload` is the human-authored new content, while `Trace` is the compatibility debt forced by sealing against the old universe. The two routes are the two composites from `El (FlatU (Arr A B))` to `Flat (El A) -> Flat (El B)`. The marginal depth-3 obligation is not the space of fillers for a closed boundary. It is the horn-extension type whose free component is the remote comparison between those composites and whose second component is the 3-cell certifying compatibility with the displayed boundary. In a cubical presentation this total extension type is canonically contractible: `hcomp` computes the missing face and `hfill` packages the accompanying filler. The extra factor is therefore present but contractible, rather than absent.

Remark 4.11 (Toy separation of d_{obl} and d_μ)**.** The cubical sketch above isolates the separation direction relevant for this paper. Fix $A, B : U$ and let $\partial(A, B)$ be the lower-arity boundary built from `flat-El`, `flat-code-Arr`, and `flat-Arr`. Write A_∂, φ_∂, u_∂, and $u_{0,\partial}$ for the cubical boundary data determined by $\partial(A, B)$. The exact depth-3 obligation over that boundary is

$$H_3(\partial(A, B)) := \sum_{\rho : A_\partial[\varphi_\partial \mapsto u_\partial(i1)]} \mathrm{PathP}(\lambda i.\, A_\partial[\varphi_\partial \mapsto u_\partial(i)], u_{0,\partial}, \rho)\,,$$

whose first component is the would-be remote comparison field `remote-face`. Cubical composition makes $H_3(\partial(A, B))$ canonically contractible, so this higher obligation is still present in $\mathcal{O}^{(3)}(X)$ even though it contributes no new field to the μ-primitive trace part N^μ_{tr}. In counting terms, the depth-3 step adds an exact contractible horn factor but zero new units to μ. This is why 4.15 and 4.13 are stated separately: even when the two depth bounds both land at 2, they measure different layers of structure.

Theorem 4.12 (Higher structural traces are transparently derived)**.** *Let e be an admissible typed raw surface extension declaration over $\mathcal{B}_n$. Let*

$$\phi \in \widehat{N}_{\mathrm{tr}}(e)$$

be a normalized structural trace field of historical arity $k \geq 3$. Then

$$\mathsf{cost}(\phi) = \mathsf{derived}.$$

Equivalently,

$$\phi \notin N^\mu_{\mathrm{tr}}(e).$$

Proof. By 3.5, ϕ is the normalized image of a typed raw structural trace clause of e. Since its arity is at least 3, the raw structural grammar of 2.2 forces that clause to have the form

$$\mathsf{horn}(\partial)$$

for an already visible lower-arity boundary ∂.

By 4.9, the normalized image of this clause is a based horn-extension package

$$H_k(\partial) = \sum_{\gamma : A_\partial[\varphi_\partial \mapsto u_\partial(i1)]} \mathrm{PathP}(\lambda i.\, A_\partial[\varphi_\partial \mapsto u_\partial(i)], u_{0,\partial}, \gamma)\,.$$

The boundary ∂ is built only from payload fields and unary or binary trace fields of the intervening sealed layers, because opacity prevents later sealing from inspecting the hidden higher derivations of those layers.

The ambient cubical core provides a canonical solution of this horn-extension problem. After any necessary transport alignment, homogeneous composition computes the missing endpoint and the associated filling operation supplies the path component:

$$\gamma_0 := \mathrm{hcomp}(\partial), \qquad \mathsf{fill}_0 := \mathsf{hfill}(\partial).$$

Thus the center $(\gamma_0, \mathsf{fill}_0) : H_k(\partial)$ is synthesized uniformly from lower-arity boundary data using only transparent cubical operations. Therefore every normalized atomic component of the arity-k structural trace package is transparently reconstructible from lower-arity data. By the cost-tag definition in 3.5, its primitive-cost tag is $\mathsf{derived}$. Hence it is not a member of $N^\mu_{\mathrm{tr}}(e)$, which contains exactly the fields tagged $\mathsf{requiresPrimitive}$. □

Theorem 4.13 (Higher-coherence elimination from μ-minimal signatures)**.** *Let e be an admissible typed raw surface extension declaration over $\mathcal{B}_n$. Then there exists a presentation-equivalent declaration*

$$e^\mu \approx e$$

whose μ-primitive trace part contains no historical trace field of arity $k \geq 3$:

$$\forall \phi \in N^\mu_{\mathrm{tr}}(e^\mu), \qquad \mathrm{arity}(\phi) \leq 2.$$

Equivalently, every canonical μ-minimal normalized public signature contains only unary and binary primitive historical trace fields. This is an elimination statement about the μ-primitive part, not a claim that the corresponding exact obligation types are absent from $\mathcal{O}^{(k)}(X)$ or that the full trace presentation $\widehat{N}_{\mathrm{tr}}(e)$ contains no transparent higher aliases.

Proof. Start with the full canonical trace presentation $\widehat{N}_{\mathrm{tr}}(e)$. Since the typed raw surface declaration is finite, there are only finitely many normalized trace fields of arity at least 3.

Let ϕ be one such field. By 4.12,

$$\mathsf{cost}(\phi) = \mathsf{derived}.$$

By 3.8, deleting ϕ from the opaque primitive telescope and replacing any public name for it by its transparent synthesizer produces a presentation-equivalent declaration. Repeating this finite replacement process for every arity-$k \geq 3$ structural trace field gives a declaration e^μ with $e^\mu \approx e$. Its full canonical trace presentation may still contain transparent aliases for the higher fields, but its μ-primitive part contains none of them.

It remains to see that this representative is minimal. By 3.5, presentation equivalence preserves the canonical normalized signature and primitive-cost tags. Thus no presentation-equivalent declaration can remove a field tagged $\mathsf{requiresPrimitive}$ without changing the sealed public typing judgment. Hence the minimum defining μ is attained exactly by the primitive-cost filtered subpresentation

$$N^\mu_{\mathrm{tr}}(e) = \{\phi \in \widehat{N}_{\mathrm{tr}}(e) \mid \mathsf{cost}(\phi) = \mathsf{requiresPrimitive}\}.$$

Since every arity-$k \geq 3$ structural trace field is tagged $\mathsf{derived}$, no such field appears in the μ-minimal primitive part. □

Corollary 4.14. *In the fixed fully coupled cubical extension calculus, the minimal-signature depth satisfies $d_\mu \leq 2$.*

Proof. 4.13 shows that every admissible typed raw surface extension declaration is presentation-equivalent to one whose μ-primitive trace part contains no historical field of arity $k \geq 3$. This is exactly the defining condition for $d_\mu \leq 2$ in 4.6. □

4.4.1. *Exact stabilization above depth* 2.

Theorem 4.15 (Exact depth-two stabilization of obligation sets)**.** *In the fixed fully coupled cubical extension calculus, for every foundational core candidate X and every $k \geq 2$ there is a canonical equivalence*

$$\mathcal{O}^{(k)}(X) \simeq \mathcal{O}^{(2)}(X).$$

Hence the obligation depth is at most 2.

Proof. The case $k = 2$ is tautological. For $k \geq 3$, apply 4.9 to any surface representative of the candidate class of X; if X is already given abstractly, this means applying the same normal form to its normalized obligation presentation. Counting normalization makes the representative choice irrelevant. We obtain

$$\mathcal{O}^{(k)}(X) \simeq \sum_{\partial:\mathcal{O}^{(k-1)}(X)} H_k(\partial).$$

For each boundary ∂, the fiber is the based dependent-path singleton

$$H_k(\partial) = \sum_{\gamma:A_\partial[\varphi_\partial \mapsto u_\partial(i1)]} \mathrm{PathP}(\lambda i.\, A_\partial[\varphi_\partial \mapsto u_\partial(i)], u_{0,\partial}, \gamma)\,.$$

It is contractible for the standard total-space-of-paths reason. Cubical composition and filling give the canonical center: after any necessary transp alignment, hcomp computes the endpoint γ, and the corresponding hfill gives the path component [CCHM18, CHM18]. Hence the projection

$$\sum_{\partial:\mathcal{O}^{(k-1)}(X)} H_k(\partial) \to \mathcal{O}^{(k-1)}(X)$$

is a canonical equivalence. Thus $\mathcal{O}^{(k)}(X) \simeq \mathcal{O}^{(k-1)}(X)$ for every $k \geq 3$, and iterating yields $\mathcal{O}^{(k)}(X) \simeq \mathcal{O}^{(2)}(X)$ for all $k \geq 2$.

In the taxonomy of 4.2, the higher exact obligation types remain present but contractible at the metatheoretic level, and only the surface-to-horn bridge together with 3.7 turns that fact into elimination from μ-minimal syntax. □

Corollary 4.16 (Contractible-factor decomposition)**.** *For every foundational core candidate X and every $k \geq 2$ there is a canonical equivalence*

$$\mathcal{O}^{(k)}(X) \simeq \mathcal{O}^{(2)}(X) \times C_k(X),$$

where $C_2(X) := \mathbf{1}$ and, for $k > 2$, $C_k(X)$ is the iterated chain of canonical remote horn factors selected by the cubical compositions used in the proof of 4.15. Each $C_k(X)$ is canonically contractible. Equivalently, the projection to the depth-two boundary trivializes the higher exact data as a contractible remote factor: the extra higher layer remains present in the exact obligation set, but the quotient that identifies each remote horn chain with its canonical center recovers precisely $\mathcal{O}^{(2)}(X)$.

Proof. The proof of 4.15 is iterative: each passage from depth j to depth $j+1$ adjoins one horn-extension fiber over already exported boundary data, and cubical composition supplies a canonical center of that fiber. Iterating those chosen centers trivializes the resulting dependent family into a product of the depth-two boundary with a remote-factor chain $C_k(X)$. Each stage of that chain is contractible by the same Kan-composition argument used in 4.15, so the whole factor $C_k(X)$ is canonically contractible. Projecting away that factor yields the displayed equivalence, and collapsing each factor to its chosen center gives the stated quotient description. □

Theorem 4.17 (Recent-history factorization)**.** *In the fixed fully coupled cubical extension calculus, every binary trace field that survives in the μ-primitive part of a μ-minimal normalized public signature at stage $n+1$ factors through the previous two normalized signatures. More precisely, let X be a candidate added at stage $n+1$. Once the unary and binary trace data of X against L_n and L_{n-1} have been fixed, every comparison between X and a remote layer L_{n-r} with $r \geq 2$ is definitionally equal to a term obtained by iterating the factorization*

through the already exported unary bridges and binary comparison traces of the intervening normalized signatures. Consequently no μ-minimal public signature contains a primitive remote binary field. This is a chronological factorization statement about surviving primitive fields, not an exact claim that remote binary obligation types are absent from $\mathcal{O}^{(2)}(X)$.

Proof. Start with the first nonadjacent layer L_{n-2}. By 3.12 applied to the sealed exports of L_n and L_{n-1}, the next factorization step may be read entirely from explicit public trace data. In particular, the export of L_n includes the adjacent unary bridge

$$f_n : L_n \to L_{n-1},$$

the export of L_{n-1} includes the adjacent unary bridge

$$g_{n-1} : L_{n-1} \to L_{n-2},$$

and the binary part of the trace of L_n contains a 2-cell

$$\alpha_n : g_{n-1} \circ f_n \Rightarrow h_n$$

recording the induced comparison $h_n : L_n \to L_{n-2}$. The degenerate composition faces needed to assemble the next open box are part of the same factorization-complete export, in the sense of 3.11, because counting normalization treats them as transparent structural data rather than as fresh primitive fields. When X is sealed against the last two layers, it additionally fixes unary comparisons

$$x_n : X \to L_n, \qquad x_{n-1} : X \to L_{n-1},$$

and a binary coherence square

$$\beta_n : f_n \circ x_n \Rightarrow x_{n-1}.$$

The cells x_n, x_{n-1}, f_n, g_{n-1}, α_n, and β_n, together with the evident degenerate composition faces, determine an open 3-box. The marginal obligation is not merely a bare remote face; it is the horn-extension type

$$H_{n-2} := \sum_{\gamma_{n-2} : h_n \circ x_n \Rightarrow g_{n-1} \circ x_{n-1}} \mathrm{PathP}(\lambda i.\, A_{n-2}[\varphi_{n-2} \mapsto u_{n-2}(i)], u_{0,n-2}, \gamma_{n-2}),$$

where A_{n-2}, φ_{n-2}, u_{n-2}, and $u_{0,n-2}$ are the boundary type, visible face formula, partial element, and lower cap determined by the displayed traces. Its first component is precisely the comparison

$$\gamma_{n-2} : h_n \circ x_n \Rightarrow g_{n-1} \circ x_{n-1}$$

between the two composites from X to L_{n-2}, and its second component is the based path filling the box once that face is chosen. Cubical Kan filling computes this horn-extension term from the displayed boundary data: hcomp synthesizes the missing face and hfill the accompanying filler. Thus the comparison with L_{n-2} factors through the already exported signature of L_n together with the adjacent bridge carried by L_{n-1}; it does not survive as a new primitive remote field in a μ-minimal presentation.

The same argument now iterates. Suppose inductively that the comparison from X to L_{n-r+1} has already been factored through the recent history. Apply 3.12 to the sealed layer L_{n-r+1}. Its normalized public trace export already supplies the adjacent unary bridge to L_{n-r}, the corresponding binary comparison trace for the composite through L_{n-r}, and the transparently generated degenerate faces needed to assemble the next open 3-box. Combining that explicit exported data with the already factored comparison for X yields another horn-extension fiber over visible boundary data. Cubical Kan filling computes that fiber

again, yielding a comparison from X to L_{n-r} which factors through the normalized public signature of L_{n-r+1}. Inductively, every remote binary comparison is therefore a definitional consequence of the trace data already exported by the recent history, and 4.13 removes it from any μ-minimal signature as a primitive field. □

Corollary 4.18 (Two-layer chronological window)**.** *In the fixed fully coupled cubical extension calculus, every sealing step at stage $n+1$ has a genuine chronological window of size 2: all primitive obligations are generated by the exported traces of L_n and L_{n-1}, while comparisons with layers L_{n-r} for $r \geq 2$ factor through those two normalized signatures and are then synthesized by iterated Kan filling. Equivalently, the pair (L_n, L_{n-1}) forms a chronological Markov blanket for subsequent sealing.*

Proof. 4.15 shows the exact obligation-set stabilization $\mathcal{O}^{(k)}(X) \simeq \mathcal{O}^{(2)}(X)$ for all $k \geq 2$. 4.17 upgrades that dimensional statement to a temporal one: every surviving binary clause factors through the normalized signatures of the last two layers, so remote history contributes only boundary data already mediated by those exports. Hence primitive sealing data are exhausted by $S(L_n) \uplus S(L_{n-1})$, and all deeper history contributes only derived fillers. □

4.4.2. *Lower bound: explicit binary obstruction ($d \geq 2$).*

Theorem 4.19 (Explicit binary sealing obstruction)**.** *Let the active interface of a fully coupled univalent foundation export the two-point type* $\mathbf{2}$ *and the univalent path*

$$p_{\mathsf{swap}} := \mathsf{ua}(\mathsf{swap}) : \mathbf{2} \equiv \mathbf{2}.$$

Then there exists a globally acting promoted interface candidate whose unary clauses do not determine its sealing: sealing induces a genuine binary obligation over p_{swap}. Consequently the obligation depth is at least 2.

Proof. Consider the promoted interface candidate X whose advertised payload is a global endomap package

$$f : \Pi_{A:\mathcal{U}} A \to A.$$

At the current sealing stage, primitive clauses need only be supplied against the active exported interface. To specify one unary clause of this advertised action, choose

$$f_{\mathbf{2}} := \lambda_.\mathsf{left} : \mathbf{2} \to \mathbf{2}.$$

Because X is advertised as globally acting, sealing it against the active interface requires coherence not only on objects but also along exported paths in $\mathcal{U}$. The path p_{swap} therefore induces the binary obligation

$$\alpha_{\mathsf{swap}} : \mathsf{transport}_{\lambda A.\, A \to A}(p_{\mathsf{swap}}, f_{\mathbf{2}}) \equiv f_{\mathbf{2}}.$$

Univalence computes transport in the endomorphism family by conjugation with swap, so

$$\mathsf{transport}_{\lambda A.\, A \to A}(p_{\mathsf{swap}}, f_{\mathbf{2}}) = \mathsf{swap} \circ f_{\mathbf{2}} \circ \mathsf{swap}^{-1} = \lambda_.\mathsf{right}.$$

Hence α_{swap} is exactly the type

$$(\lambda_.\mathsf{right}) \equiv (\lambda_.\mathsf{left}),$$

which is impossible because evaluation at left would force $\mathsf{right} = \mathsf{left}$. Thus the passage from depth 1 to depth 2 introduces a genuine new sealing obligation type: the unary clause $f_{\mathbf{2}}$ does not determine whether X is admissible, because the typechecker must still ask for the binary witness α_{swap}.

Therefore $\mathcal{O}^{(2)}(X)$ contains a binary obligation not already present in $\mathcal{O}^{(1)}(X)$, so the ambient foundation cannot have obligation depth 1. Equivalently, univalence exposes binary sealing sites that are invisible to unary data alone. This is the same obstruction formalized in `Metatheory/AdjunctionBarrier.agda` by the nontriviality of `swap-path` and the contradiction theorem `depth1-insufficient`. Hence $d_{\mathrm{obl}} \geq 2$. □

Remark 4.20 (Adjunction as categorical reading)**.** The triangle identities of an adjunction exhibit the same binary pattern at a categorical level: once the object part and the unit/counit data have been fixed, the remaining triangle coherence compares composites already determined at arity 1. The paper uses adjunctions in that explanatory role only. The internal lower-bound theorem of the extension calculus is the swap obstruction above.

4.4.3. *Topological exact family: clutching constructions.*

Theorem 4.21 (Clutching binary lower-bound family)**.** *The HIT clutching presentation over S^2 furnishes a depth-two lower-bound witness: a genuine binary comparison is necessary to specify the bundle data, so unary data alone do not determine admissibility.*

Proof. We translate the clutching construction into higher-inductive syntax. In homotopy type theory, the 2-sphere is the suspension $S^2 := \Sigma S^1$, generated by point constructors

$$\mathsf{N}, \mathsf{S} : S^2$$

and a path constructor

$$\mathsf{merid} : S^1 \to (\mathsf{N} \equiv \mathsf{S})$$

[Uni13, CHM18]. To specify an exported layer whose total space is a principal bundle over S^2, it suffices to define a type family

$$P : S^2 \to \mathcal{U}.$$

By the dependent eliminator for the suspension, such a family is determined by exactly three pieces of data:

$$P_{\mathsf{N}} : \mathcal{U}, \qquad P_{\mathsf{S}} : \mathcal{U}, \qquad \gamma : \Pi_{x:S^1}\big(P_{\mathsf{N}} \equiv P_{\mathsf{S}}\big).$$

By univalence, the last component may equally be read as a family of equivalences between the two pole fibers, which is the type-theoretic clutching datum.

Now S^1 is itself a higher inductive type generated by a point base and a loop $\mathsf{loop} : \mathsf{base} \equiv \mathsf{base}$. Therefore giving γ requires exactly two layers of information: first a base equivalence

$$\gamma(\mathsf{base}) : P_{\mathsf{N}} \equiv P_{\mathsf{S}},$$

which is unary data, and second a path over loop identifying the two transports induced by that equivalence around the circle. That second datum is a genuine binary coherence cell: it compares two unary composites and cannot be recovered from the endpoints alone. Hence depth 1 does not suffice.

Thus the clutching presentation exhibits a genuine binary datum that is not recoverable from unary endpoints, furnishing the stated lower-bound witness. □

Remark 4.22 (Role of clutching in the depth theorem)**.** 4.21 is used only for exactness of the lower bound: it rules out depth 1 by exhibiting nontrivial binary data. It is not an upper-bound argument. The statement that independent arity-3 structural integration data do not survive in the fixed cubical extension calculus is supplied separately by 4.9, 4.15,

and 4.13. In the clutching example itself, the HIT signature of S^2 has no constructor of dimension 3 or higher; this is compatible with, but does not prove, the general upper-bound theorem.

Corollary 4.23. *In the fixed fully coupled cubical extension calculus considered above, the obligation depth is exactly* 2.

Proof. 4.15 provides the exact stabilization statement by showing the obligation-set equivalence $\mathcal{O}^{(k)}(X) \simeq \mathcal{O}^{(2)}(X)$ for all $k \geq 2$, while 4.13 gives the separate syntactic statement that no μ-minimal public signature needs primitive arity-3 trace data. 4.17 shows that this depth-two exact-stabilization statement is realized by a genuine two-layer chronological window rather than merely by a bound on cell dimension. 4.19 and 4.21 provide lower bounds by exhibiting genuine depth-two phenomena that do not collapse in the univalent setting. Therefore the least stabilizing obligation depth is 2. □

5. The complexity scaling theorem

Let e_n be the admissible typed raw surface extension declaration whose sealing exports the nth sealed layer L_n, and write

$$\mu_n := \mu(e_n)$$

for its minimal opaque cost, and let κ_n be its core payload size. By 3.10, μ_n is exactly the abstract sealing cost $\Delta(X(e_n) \mid \mathcal{B}_{n-1})$, but it is read from the μ-primitive filter of a full tagged canonical public signature and is therefore presentation-invariant by construction. The recurrence theorem is applied below together with the obligation-depth statement of 4.6, but its formal counted-interface package only needs a theorem identifying a genuine chronological window. In the depth-two cubical univalent case, this is exactly 4.18; in the UIP case, the proof of 4.8 shows that all non-unary comparisons collapse to their boundary, so the window is one layer. More generally, whenever a foundation has obligation depth d_{obl} realized by a chronological window of size d_{obl}, any depth-d_{obl} historical basis of the preceding layers has the form

$$I_{n,\mathrm{bas}}^{(d_{\mathrm{obl}})} = \biguplus_{j=0}^{d_{\mathrm{obl}}-1} S_{\mathrm{bas}}(L_{n-j}).$$

By 2.19, its cardinality is independent of the basis families used to write the coproduct. Thus the recurrence counts the well-defined irreducible basis cardinality of the active interface rather than the raw number of exported generators. For the initial bootstrap stages before the full window has been populated, we adopt the natural convention that nonexistent earlier layers contribute the empty interface.

5.1. Sparse footprints and the full-coupling envelope.

Before specializing to fully coupled foundations, it is useful to record the sparse form of the counting law. A sealed extension may advertise action only on a proper dependency footprint inside the recent historical interface. This is the ordinary case for local datatypes, APIs, or plugins: they can be opaque and still interact with only the layers they mention.

Definition 5.1 (Sparse coupling footprint)**.** Assume a foundation has a chronological window of size d. A *coupling footprint* for the next sealed layer is a finite subset

$$F_n \subseteq \{0, \dots, d-1\}$$

whose element j means that the extension is advertised as acting on the basis sites of L_{n-j}. The active basis for that extension is then

$$I^{F_n}_{n,\mathrm{bas}} := \biguplus_{j \in F_n} S_{\mathrm{bas}}(L_{n-j}).$$

The fully coupled recent-history case is $F_n = \{0, \dots, d-1\}$.

Theorem 5.2 (Sparse footprint recurrence)**.** *For a sparse sealed extension discipline whose sealing steps satisfy the active-basis contract on their advertised footprints and whose primitive obligations factor through a chronological window of size d, the minimal opaque cost satisfies*

$$\mu_{n+1} = \sum_{j \in F_n} \big(\mu_{n-j} + \kappa_{n-j}\big).$$

Consequently the fully coupled recurrence of 5.3 is the special case $F_n = \{0, \dots, d_{\mathrm{obl}} - 1\}$, and the cubical Fibonacci law of 5.5 is the depth-two, constant-payload, complete-footprint specialization.

Proof. The proof is the same counting argument as 5.3, restricted to the advertised footprint. By 2.26, applied only to the basis sites in $I^{F_n}_{n,\mathrm{bas}}$, the next minimal opaque cost is the cardinality of that footprint basis. The tagged coproduct decomposition gives

$$|I^{F_n}_{n,\mathrm{bas}}| = \sum_{j \in F_n} |S_{\mathrm{bas}}(L_{n-j})|.$$

For each included layer, 3.10 and the trace principle identify the layer's basis size with $\mu_{n-j} + \kappa_{n-j}$. Substitution gives the sparse recurrence. Taking the complete recent-history footprint recovers 5.3; taking $d = 2$ and uniform payload size recovers 5.5. □

This sparse theorem is the reason orthogonal examples are not counterexamples to the paper's main law. They live below the full-coupling envelope because their advertised footprint omits unrelated layers. The Fibonacci recurrence describes the maximal cubical foundational-core endpoint, not arbitrary repository growth.

5.2. **The universal affine recurrence.**

Theorem 5.3 (Universal affine recurrence for minimal opaque cost)**.** *For every fully coupled foundation whose sealing steps satisfy the active-basis contract and whose obligation depth d_{obl} is realized by a chronological window of size d_{obl}, the minimal opaque cost satisfies*

$$\mu_{n+1} = \sum_{j=0}^{d_{\mathrm{obl}}-1} \big(\mu_{n-j} + \kappa_{n-j}\big) \qquad (n \geq d_{\mathrm{obl}}).$$

Proof. By 3.10, the next minimal opaque cost is the abstract sealing cost, so 2.26 identifies it, under the active-basis contract, with the size of the active basis interface:

$$\mu_{n+1} = |I^{(d_{\mathrm{obl}})}_{n,\mathrm{bas}}|.$$

Using the tagged coproduct decomposition of 2.18,

$$|I_{n,\mathrm{bas}}^{(d_{\mathrm{obl}})}| = \sum_{j=0}^{d_{\mathrm{obl}}-1} |S_{\mathrm{bas}}(L_{n-j})|.$$

Now apply 3.10, equivalently the Integration Trace Principle 2.27, which gives

$$|S_{\mathrm{bas}}(L_{n-j})| = \mu_{n-j} + \kappa_{n-j}$$

for each j. Substituting yields the claimed affine recurrence. □

For later comparison, write

$$\tau_n := \sum_{i=1}^{n} \mu_i$$

for the cumulative minimal opaque cost through stage n.

5.3. **Affine one-step growth at $d = 1$.**

Corollary 5.4. *For a fully coupled foundation whose sealing steps satisfy the active-basis contract, if the obligation depth is* 1*, this depth is realized by a one-layer chronological window, and every layer contributes the same core payload size c, then*

$$\mu_{n+1} = \mu_n + c.$$

If moreover the initial state exports no prior sealed layers, then

$$\mu_1 = 0, \qquad \mu_n = (n-1)c,$$

and therefore

$$\tau_n = c\,\frac{n(n-1)}{2}.$$

Minimal opaque cost therefore grows linearly. The cumulative opaque cost grows quadratically.

Proof. Substitute $d = 1$ into 5.3. Then only the term $j = 0$ remains, so

$$\mu_{n+1} = \mu_n + \kappa_n.$$

Under the uniform payload assumption $\kappa_n = c$, this becomes $\mu_{n+1} = \mu_n + c$. If the initial state exports no prior sealed layers, then the first sealing step sees the empty interface and therefore has minimal opaque cost $\mu_1 = 0$. Induction gives $\mu_n = (n-1)c$, and summing this arithmetic progression yields the stated formula for τ_n. □

5.4. **Affine Fibonacci scaling at $d = 2$.** By 4.18, the fixed cubical depth-two theorem is chronologically realized by the two most recent sealed traces.

Assume in this subsection that every sealed layer contributes the same core payload size c, and write $(F_n)_{n\geq 1}$ for the Fibonacci sequence with $F_1 = F_2 = 1$.

Corollary 5.5 (Affine Fibonacci recurrence)**.** *In the fixed fully coupled cubical extension calculus, whose admissible sealing steps satisfy the active-basis contract, if the obligation depth is* 2 *and every sealed layer contributes core payload size c, then*

$$\mu_{n+1} = \mu_n + \mu_{n-1} + 2c.$$

If we define

$$U_n := \mu_n + 2c,$$

then

$$U_{n+1} = U_n + U_{n-1}.$$

If moreover the initial state exports no prior sealed layers, then

$$\mu_1 = 0, \qquad \mu_2 = c,$$

so

$$U_n = cF_{n+2}, \qquad \mu_n = c(F_{n+2} - 2),$$

and the cumulative opaque cost satisfies

$$\tau_n = c(F_{n+4} - 2n - 3).$$

Proof. The recurrence is the case $d = 2$ of 5.3 together with the uniform payload assumption $\kappa_n = c$, giving

$$\mu_{n+1} = \mu_n + \mu_{n-1} + 2c.$$

Adding $2c$ to both sides yields

$$U_{n+1} = U_n + U_{n-1}.$$

If the initial state exports no prior sealed layers, then the first layer has no earlier interface against which to cohere, so $\mu_1 = 0$. The second layer sees only the first layer's export, whose size is $\kappa_1 + \mu_1 = c$, hence $\mu_2 = c$. Therefore

$$U_1 = 2c = cF_3, \qquad U_2 = 3c = cF_4.$$

Since both (U_n) and (cF_{n+2}) satisfy the same Fibonacci recurrence and agree at $n = 1, 2$, induction gives $U_n = cF_{n+2}$. Subtracting $2c$ yields

$$\mu_n = c(F_{n+2} - 2).$$

Summing from $i = 1$ to n gives

$$\tau_n = c\sum_{i=1}^{n}(F_{i+2} - 2) = c(F_{n+4} - 2n - 3),$$

using the standard Fibonacci summation identity. □

Corollary 5.6 (Asymptotic inflation factor)**.** *Under the bootstrap and uniform-payload hypotheses of 5.5, the structural inflation factor*

$$\Phi_n := \frac{\mu_n}{\mu_{n-1}}$$

converges to the Golden Ratio

$$\varphi = \frac{1 + \sqrt{5}}{2}.$$

Proof. Under the bootstrap assumptions of 5.5,

$$\Phi_n = \frac{\mu_n}{\mu_{n-1}} = \frac{F_{n+2} - 2}{F_{n+1} - 2}.$$

The additive constants are asymptotically negligible relative to the Fibonacci growth, so the standard ratio convergence gives $\Phi_n \to \varphi$. □

5.5. **Metatheoretic robustness and universality across 2D foundations.** The counting notions used above are not source-level line counts. They are dimensions of the sealed typing judgment after passage to the counting normal form of 2.7. The first corollary records that robustness explicitly.

Corollary 5.7 (Robustness under admissible refactoring)**.** *Let two presentations of the same sealed payload or obligation judgment be related by canonical telescope isomorphism: rebundling or splitting dependent records, reassociation of Σ-types, currying or uncurrying iterated function arguments, or transparent internal normalization that does not change which fields are primitively inhabited. Then the induced sets of atomic payload schemas, atomic obligations, and historical supports are canonically bijective. In particular, the quantities $\kappa(X)$, $\Delta(X \mid \mathcal{B})$, and $a(o)$ are unchanged, and for surface declarations the induced minimal opaque cost $\mu(e)$ is unchanged as well.*

Proof. All of the listed refactorings preserve the same dependent typing judgment and differ only by canonical rebracketing or transparent elaboration of its telescope. After flattening the Σ-spine and tagging uniformly synthesized trace fields as derived, both presentations reduce to the same counting normal form of 2.7 and 4.1. The atomic payload generators, atomic obligations, and their supporting layer indices therefore correspond bijectively. Since the paper's metrics are cardinalities of those normal-form sets, they are invariant under such refactorings. The corresponding trace subfamilies of full tagged normalized public signatures are therefore canonically bijective as well, so 3.6 gives the invariance of μ. □

The cubical depth-two theorem can now be separated into two abstract layers. The stronger one is a *horn-computational* interface: it packages exactly the hypotheses that turn cubical exact stabilization of obligation sets into the bridge statement that higher trace fields disappear from μ-minimal syntax. The weaker one keeps only the primitive/window and exact-stabilization data needed for the depth-two affine law. In this paper the stronger horn-computational package is instantiated only for the chosen CCHM-style cubical core, while the weaker exact/window wrapper also supports the explicit strict/fibrant 2LTT-style toy discipline.

Theorem 5.8 (Abstract horn-computational foundation interface)**.** *Let a fully coupled foundation satisfy the following package.*

(1) *For every candidate X, every $k \geq 3$, and every boundary $\partial : \mathcal{O}^{(k-1)}(X)$, the depth-$k$ obligation over ∂ is canonically represented by a horn-extension fiber*

$$H_k(\partial) := \sum_{\gamma : A_\partial[\varphi_\partial \mapsto u_\partial(i1)]} \mathrm{PathP}(\lambda i.\, A_\partial[\varphi_\partial \mapsto u_\partial(i)], u_{0,\partial}, \gamma)\,,$$

and there is a canonical equivalence

$$\mathcal{O}^{(k)}(X) \simeq \sum_{\partial : \mathcal{O}^{(k-1)}(X)} H_k(\partial).$$

(2) *Every horn-extension fiber $H_k(\partial)$ of Property (1) is canonically contractible.*
(3) *Every admissible typed raw surface extension declaration admits a full canonical normalized trace presentation, together with presentation equivalence and primitive-versus-derived tagging on its trace fields.*
(4) *In the canonical presentation of Property (3), every historical trace field of arity $k \geq 3$ corresponding to a horn factor of Property (1) is marked* transparently derived*, and computational replacement by the canonically synthesized horn witness preserves the full canonical trace presentation while removing the field from the μ-primitive part.*
(5) *Primitive remote binary comparisons satisfy telescopic factorization: for every $r \geq 2$, the comparison between X and L_{n-r} is obtained by one horn-extension step from the comparison between X and L_{n-r+1} together with the factorization-complete public trace export of L_{n-r+1}.*

Then the foundation satisfies all three conclusions below.

(1) *For every candidate X and every $k \geq 2$ there is a canonical equivalence*
$$\mathcal{O}^{(k)}(X) \simeq \mathcal{O}^{(2)}(X).$$
(2) *Every admissible typed raw surface extension declaration is presentation-equivalent to one whose μ-primitive trace part contains no historical field of arity $k \geq 3$.*
(3) *Primitive historical support factors through a chronological window of size 2.*

In particular, the exact obligation-depth bound $d_{\mathrm{obl}} \leq 2$ and the minimal-signature bound $d_\mu \leq 2$ both hold.

Proof. By Property (1), each passage from depth $k-1$ to depth k with $k \geq 3$ decomposes the exact obligation set into a depth-$(k-1)$ boundary together with one horn factor. Property (2) makes that horn factor contractible, so the projection to the boundary is a canonical equivalence. Iterating from depth k down to depth 2 yields
$$\mathcal{O}^{(k)}(X) \simeq \mathcal{O}^{(2)}(X)$$
for every candidate X and every $k \geq 2$, which is exactly the claimed exact stabilization above depth 2.

For the surface-syntactic conclusion, fix an admissible typed raw surface declaration e and any historical field ϕ of arity $k \geq 3$ occurring in its full canonical trace presentation. By Property (3), its contribution may be read on the full canonical trace presentation attached to e. By Property (4), the corresponding canonical field is marked transparently derived and may be replaced by the horn-computed witness without changing that full canonical presentation. Hence ϕ contributes no requires-primitive trace schema in any μ-minimal representative. Repeating this finite replacement for each arity-$k \geq 3$ field yields a presentation-equivalent declaration whose μ-primitive trace part has no historical field above arity 2. This is exactly the asserted elimination statement, and by 4.6 it gives $d_\mu \leq 2$.

It remains to derive the chronological window. After the preceding paragraph, every primitive field in a μ-minimal canonical trace presentation is unary or binary. Unary fields are supported on the adjacent public interface. For binary fields, Property (5) factors every remote comparison one layer at a time through the previous comparison and the factorization-complete trace export of the intervening sealed layer. Iterating that telescopic factorization rewrites every remote binary primitive into data mediated by the two most recent normalized signatures. Thus primitive historical support factors through a chronological window of size 2. □

Corollary 5.9 (Cubical instantiation of the horn-computational interface)**.** *The fixed fully coupled cubical extension calculus over the chosen CCHM-style core satisfies the package of 5.8. Consequently 4.15, 4.13, and 4.18 are the concrete instance of that abstract horn-computational theorem pattern checked in this paper.*

Proof. Property (1) is exactly the surface-to-horn normal form of 4.9 together with the horn-reduction description of 4.10; Property (2) is the cubical contractibility argument used in 4.15 and 4.16; Property (3) is the canonical trace-presentation interface supplied by section 3 together with the explicit presentation interface of 3.7; Property (4) is the computational-replacement clause of 3.7; and Property (5) is the telescopic remote-comparison factorization theorem 4.17, using factorization-complete export from 3.12. Therefore 5.8 applies to the chosen cubical calculus. □

The theorem above is intentionally stronger than the exact-depth wrapper used below: it includes the canonical trace-presentation and computational-replacement clauses needed to transport horn computation all the way to elimination from μ-minimal syntax. The next theorem forgets that bridge data and keeps only the weaker primitive/window and exact-stabilization package formalized in `Metatheory/TwoDFoundations.agda`; that is the level at which the separate strict/fibrant 2LTT-style discipline is checked.

Theorem 5.10 (Depth-two law for 2D foundations)**.** *Let a fully coupled foundation whose sealing steps satisfy the active-basis contract also satisfy the following two properties.*

(1) *There exists a candidate X whose normalized binary obligation set is not canonically equivalent to its unary one:*
$$\mathcal{O}^{(2)}(X) \not\simeq \mathcal{O}^{(1)}(X).$$
(2) *For every candidate X and every $k \geq 2$, there is a canonical equivalence*
$$\mathcal{O}^{(k)}(X) \simeq \mathcal{O}^{(2)}(X),$$
and every surviving binary clause factors through a chronological window of size 2.

Then the foundation has obligation depth 2. Under the uniform payload hypothesis $\kappa_n = c$, its minimal opaque cost satisfies
$$\mu_{n+1} = \mu_n + \mu_{n-1} + 2c,$$
so the shifted sequence $U_n := \mu_n + 2c$ is Fibonacci. The hypotheses concern exact obligation stabilization together with chronology; any separate elimination theorem for μ-minimal signatures is additional structure, not part of this abstract wrapper.

Proof. Property (2) gives exact stabilization $\mathcal{O}^{(k)}(X) \simeq \mathcal{O}^{(2)}(X)$ for all $k \geq 2$ together with a genuine two-layer chronological window. Hence 4.6 and 5.3 imply that the obligation depth is at most 2 and that the recurrence is the depth-two affine law. Property (1) excludes obligation depth 1. Hence the least stabilizing obligation depth is exactly 2, and substituting $d_{\mathrm{obl}} = 2$ and $\kappa_n = c$ into 5.3 gives the displayed formula. The Fibonacci shift is exactly the argument of 5.5. □

Corollary 5.11 (A concrete 2LTT instantiation)**.** *There is a concrete strict/fibrant 2LTT-style extension discipline satisfying the hypotheses of 5.10. In that discipline, arities 0 and 1 carry no primitive obligations, arity 2 carries a primitive fibrant binary comparison, every arity $k \geq 3$ carries the same comparison schema but only as transparently derived data, and every surviving primitive support factors through the previous two layers. Consequently this*

Table 3: Scope of the paper's generalization claims.

Foundation	Binary lower bound	Exact stabilization / window	Bridge to μ	Claimed here?
Chosen CCHM-style cubical core / Cubical Agda realization with active-basis contract	yes; explicit swap and clutching witnesses	yes; horn reduction, stabilization, and recent-history factorization	yes; replacement interface formalized, surface-to-interface normalization paper-level	yes
Other cubical variants	only if a binary witness is checked case by case	only if the horn-reduction, contractible-fiber, and window clauses of 5.8 are verified	only if the canonical-trace bridge clauses of 5.8 are verified	no
Explicit strict/fibrant 2LTT-style discipline	yes; concrete primitive binary witness	yes; exact stabilization and size-two window checked	not supplied here; only the weaker wrapper 5.10 is claimed	yes, for that discipline only
Plain Book HoTT	no theorem isolated here	no cubical-style exact-stabilization argument supplied here	no	no

discipline has obligation depth 2 *and chronological window size* 2*. No claim is made here that it also satisfies the stronger horn-computational bridge package of 5.8.*

Proof. Take the explicit strict/fibrant discipline just described. Since the binary comparison is primitive while the unary layer is empty, Property (1) of 5.10 holds. For every $k \geq 2$, the obligation family is the same binary comparison schema, so the exact stabilization part of Property (2) is immediate; and because its support is fixed to the last two layers, the size-two chronological window is immediate as well. Therefore 5.10 applies. The accompanying mechanization packages this instance in `Metatheory/TwoLTTFoundations.agda` via `two-level-primitive-window-2d-foundation`, `two-level-2d-foundation`, `two-level-depth-two-law-for` and `two-level-window-size-two-for-2d-foundations`. □

Remark 5.12 (Why plain HoTT is not claimed here)**.** The identity-elimination principle of the HoTT book reduces families over a variable path to the reflexivity constructor, but it does not by itself collapse loop-indexed families with fixed endpoints to the reflexivity case [Uni13]. The horn-extension problems governing 4.10 therefore do not automatically reduce in plain Book HoTT to the cubical boundary computations used in Properties (1)–(2) of 5.8. For that reason the paper does not claim that plain HoTT instantiates 5.8, nor even that it satisfies the weaker exact/window hypothesis of Property (2) in 5.10; for this paper it remains an open direction rather than a theorem-level corollary.

For the chosen CCHM-style cubical core, 5.9 verifies the stronger horn-computational package of 5.8. The weaker theorem 5.10 deliberately forgets the canonical-trace and computational-replacement clauses and keeps only the exact/window data needed for the affine law; that is why the strict/fibrant 2LTT-style discipline of 5.11 can instantiate 5.10 without being claimed to satisfy 5.8. Plain HoTT remains outside both theorem-level instantiations in this paper.

6. Cubical Agda mechanization

6.1. **Mechanizing the theorem package.** The formalization is organized around the fixed extension calculus of section 2, its horn-computational abstraction layer, its depth theorems, the induced recurrence law, and the clutching construction. Table 4 first makes the trusted boundary local. Here *fully formalized* means that the paper's statement is realized directly in the cited Agda modules; *formalized abstractly, read through bridge* means that the Agda development proves the structural theorem and the paper reads it through the normalization and canonical-presentation results of section 3; and *paper-level only* marks the bridge layer itself. In particular, 3.7 is fully formalized on explicit canonical trace presentations. The paper-level part of the bridge is the surface-to-interface normalization package of section 3, including the surface-to-horn image theorem 4.9: raw structural declarations are first normalized into the canonical horn-generated obligation language, and only then are the formal horn theorems applied. The itemized map after the table records the exact paper-to-Agda theorem correspondence.

If this bridge is pushed one step further into Agda, the intended object is a typed bridge contract rather than a parser theorem. A module such as `Metatheory/SurfaceNormalizationBridge.agda` would package a raw declaration type, its elaborated candidate, raw exported trace fields at each depth, the associated `CanonicalTraceSignature` and `TracePresentation`, an isomorphism from raw fields to presented canonical fields, presentation-equivalence soundness and completeness, and the statement that every raw structural trace field of depth at least 3 maps to a presented field whose primitive cost is `derived`. The paper's 3.5, 4.12, and 4.13 are exactly this contract read in mathematical form.

- 2.14 corresponds exactly to `Metatheory/InterfaceCalculus.agda` via `transparent-growth-keeps-library-state`, `transparent-definitions-preserve-active-interface`, `transparent-definitions-have-zero-integration-l` and `transparent-user-level-code-lies-outside-the-recurrence`.
- 2.19 is a paper-level finiteness and canonical-bijection theorem for normal-form basis representatives: it packages counting normalization and 3.6 into the claim that every basis family picks one representative from each irreducible transparent-generation class. It is not a uniqueness theorem for arbitrary finite closure systems. The Agda development does not separately formalize these representative choices; instead it takes the counted active interface as primitive.
- 2.26 is the paper's active-basis-density reformulation of `Metatheory/CanonicityDensity.agda` via `CanonicityDensityTheorem` and `global-admissibility-forces-maximal-interface-density`, combined with the bridge-level normalization and canonical-presentation results of section 3. That Agda module formalizes the abstract pattern that explicit totality or exhaustiveness fields supply one primitive interaction per generator; it is not an independent operational canonicity proof for CCHM or for arbitrary Cubical Agda programs.

Table 4: Mechanization status of the main theorem stack.

Paper result	Formal status	Agda module(s)	Bridge?
section 3 normalization package and 4.9	paper-level only	none	n/a
3.7	fully formalized on explicit canonical trace presentations	`Metatheory/ComputationalReplacement.agda`	yes
2.14	fully formalized	`Metatheory/InterfaceCalculus.agda`	no
2.19	paper-level only	none	n/a
2.26, 2.27, 3.10	contract-to-density formalized abstractly, read through bridge	`Metatheory/CanonicityDensity.agda`, `Metatheory/TracePrinciple.agda`, `Metatheory/InterfaceCalculus.agda`	yes
4.5, 4.8	fully formalized	`Metatheory/Obligations.agda`, `Metatheory/Extensional.agda`	no
4.10, 4.15, 4.16, 4.18	formalized for the horn obligation language	`Metatheory/KanSubsumption.agda`, `Metatheory/UpperBound.agda`, `Metatheory/ChronologicalWindow.agda`	bridge
3.12, 4.13, 4.17	formalized abstractly, with the replacement step itself formalized	`Metatheory/KanSubsumption.agda`, `Metatheory/UpperBound.agda`, `Metatheory/ChronologicalWindow.agda`, `Metatheory/ComputationalReplacement.agda`	yes
5.8, 5.9	formalized abstractly, read through bridge	`Metatheory/KanSubsumption.agda`, `Metatheory/UpperBound.agda`, `Metatheory/ChronologicalWindow.agda`, `Metatheory/ComputationalReplacement.agda`	yes
4.19, 4.21, 4.23	fully formalized	`Metatheory/AdjunctionBarrier.agda`, `Geometry/Clutching.agda`, `Metatheory/ExactDepth.agda`	no
5.10, 5.11	fully formalized	`Metatheory/TwoDFoundations.agda`, `Metatheory/TwoLTTFoundations.agda`	no
5.2	paper-level sparse-footprint generalization of the counted recurrence	planned `Metatheory/SparseDependencyRecurrence.agda`	yes
5.3, 5.4, 5.5	formalized abstractly, read through bridge	`Metatheory/UniversalRecurrence.agda`, `Core/DepthOneAffine.agda`, `Core/AffineRecurrence.agda`	yes
5.7	fully formalized	`Metatheory/Refactoring.agda`	no

- 2.27 corresponds exactly to `Metatheory/TracePrinciple.agda` via `IntegrationTracePrinciple`, `public-counting-normal-form`, and `integration-trace-principle`, built on the explicit sealed-layer presentation `ExplicitSealedLayer` from `Metatheory/InterfaceCalculus.agda`.
- 3.10 is the paper's bridge corollary combining the active-basis contract hypothesis of 2.26, 2.27, and the normalization/canonical-presentation results of section 3; the Agda development formalizes the abstract counting theorems, while the passage to the syntactic quantity μ is the paper's normalized-signature reading of those results.
- 4.5 corresponds exactly to `Metatheory/Obligations.agda` via `historical-arity-forces-cell-dimension` and `irreducible-obligation-requires-cell`.
- 4.8 corresponds exactly to `Metatheory/Extensional.agda` via `UIP-forces-depth-1` and `history-truncates-to-on`
- 4.9 is the paper-level image theorem for the fixed extension calculus: raw structural obligations normalize into the horn-generated structural obligation language. The formal target of this bridge

is the `StructuralObligation` and `HornExtensionFiber` surface of `Metatheory/KanSubsumption.agda`; the current repository does not mechanize a parser for arbitrary surface declarations.

- 4.10 corresponds to `Metatheory/KanSubsumption.agda` via `HornExtensionFiber`, `structural-integration-horn-r` and `remote-layer-obligation-derived`, after applying the image theorem 4.9 to raw surface obligations.
- 3.7 is formalized in `Metatheory/ComputationalReplacement.agda`. That module makes the canonical trace-signature interface explicit via `CanonicalTraceSignature` and `TracePresentation`, packages the presentation theorem itself as `computational-replacement-preserves-canonical-presentation`, and exposes the higher-arity elimination consequence via `higher-arity-fields-disappear-from-minimal-signatur` and `higher-arity-computational-replacement`. The remaining paper-level ingredient is 3.5: the passage from typed raw surface declarations to that canonical interface, with primitive-cost tags preserved.
- 3.12 is the paper's bridge-level export theorem: `TelescopicSubsumptionView` in `Metatheory/KanSubsumption.agda` packages the recent boundary together with the iterated trace chain, while `primitive-obligations-factor-through-last-two` in `Metatheory/ChronologicalWindow.agda` records that primitive support factors through the last two layers. The paper theorem uses 2.27, 3.3, and 3.6 to read that formal package as an explicit normalized exported trace basis.
- 4.13 is the paper's surface-syntactic corollary of 4.9, `Metatheory/KanSubsumption.agda`, `Metatheory/UpperBound.agda`, `Metatheory/TwoDFoundations.agda`, and `Metatheory/ComputationalReplacement.agda`: the Agda development formalizes the horn-extension fiber, its canonical inhabitant, stabilization at depth 2, the abstract primitive-elimination law, and the canonical-trace replacement interface via `HornExtensionFiber`, `structural-integration-horn-reduction`, `remote-layer-obligation-derived`, `structural-obligation-set-equivalence`, `structural-stabilizes-at-two`, `structural-primitive-eliminate` `higher-arity-fields-disappear-from-minimal-signature`, and `higher-arity-computational-replacement`; the remaining paper step is the bridge from a raw surface declaration to the canonical horn and trace presentations on which those formal theorems are stated.
- 4.17 corresponds exactly to `Metatheory/KanSubsumption.agda` via `TelescopicSubsumptionView`, `telescopic-subsumption`, and `telescopic-remote-comparison-derived`; its reading as recent-history factorization through normalized public signatures is again the paper's bridge-level interpretation of that formal trace chain.
- 5.8 is the paper's stronger horn-computational wrapper. It combines the surface-to-horn image theorem 4.9, the horn-reduction and contractible-factor package from `Metatheory/KanSubsumption.agda` and `Metatheory/UpperBound.agda`, the telescopic factorization package `telescopic-subsumption` and `telescopic-remote-comparison-derived`, and the canonical-trace replacement interface of `Metatheory/ComputationalReplacement.agda` via `CanonicalTraceSignature`, `TracePresentation`, `higher-arity-fields-disappear-from-minimal-signature`, and `higher-arity-computational-replacement`, read through the bridge-level normalization package of section 3. 5.9 is the paper's checked instantiation of that wrapper for the chosen cubical calculus.
- 4.15 corresponds exactly to `Metatheory/UpperBound.agda` via `structural-obligation-set-equivalence` and `structural-stabilizes-at-two`.
- 4.16 corresponds exactly to `Metatheory/UpperBound.agda` via `ContractibleRemoteFactor`, `structural-obligation-contractible-factorization`, and `contractible-remote-factor-contractible`; this is the formal trivialization of each higher exact obligation set into the depth-two boundary together with a canonically contractible remote horn chain.
- 4.18 corresponds exactly to `Metatheory/ChronologicalWindow.agda` via `primitive-obligations-factor-through-` `two-layer-chronological-window`, and `chronological-markov-blanket`.
- 4.19 corresponds exactly to `Metatheory/AdjunctionBarrier.agda` via `explicit-binary-sealing-obstruction`.
- 4.20 corresponds to the same module's categorical wrapper via `triangle-identity-corollary` and `adjunction-barrier`.
- 4.21 corresponds exactly to `Geometry/Clutching.agda` via `clutching-family-theorem`, `hopf-binary-clutching-n` and `clutching-horn-extension-fiber-contractible`.

- 4.23 corresponds exactly to `Metatheory/ExactDepth.agda` via `structural-coherence-depth-exactly-two`, `structural-chronological-window-size-exactly-two`, `cubical-coherence-depth-exactly-two`, and `cubical-chronological-window-size-exactly-two`.
- 5.10 corresponds exactly to `Metatheory/TwoDFoundations.agda`. The weaker primitive/window layer is packaged there by `PrimitiveWindow2DFoundation`, `primitive-depth-two-law-for-2d-foundations`, and `chronological-window-size-two-for-2d-foundations`; the separate exact-depth wrapper of the theorem is packaged by `FullyCoupled2DFoundation`, `depth-two-law-for-2d-foundations`, and `constant-payload-depth-two-law`. The cubical instance is exposed there by `cubical-primitive-window-2d-fo` `cubical-primitive-depth-two-law-for-2d-foundations`, and `cubical-depth-two-law-for-2d-foundations`. The explicit strict/fibrant 2LTT-style instantiation of 5.11 is packaged in `Metatheory/TwoLTTFoundations.agda` via `two-level-language`, `two-level-primitive-window-2d-foundation`, `two-level-primitive-depth-two-` `two-level-2d-foundation`, `two-level-depth-two-law-for-2d-foundations`, and `two-level-window-size-two-`
- 5.2 is a paper-level sparse-footprint generalization of the counted recurrence. It restricts the active-basis and trace-principle count to an advertised footprint F_n and recovers 5.3 at the complete recent-history footprint. A separate Agda module `Metatheory/SparseDependencyRecurrence.agda` remains to be added.
- 5.3 corresponds exactly to `Metatheory/UniversalRecurrence.agda` via `WindowedRecurrenceContext`, `ChronologicalRecurrenceContext`, `UniversalAffineRecurrence`, `universal-affine-recurrence`, and `universal-affine-recurrence-from-coherence`, read through 3.10 so that the paper's surface-syntactic recurrence is stated over μ.
- 5.4 corresponds exactly to `Core/DepthOneAffine.agda` via `depth1-affine-growth`, `Delta-depth1-closed`, and `tau-depth1-closed`; in the paper these arithmetic laws are read through 3.10 as statements about minimal opaque cost.
- 5.5 corresponds exactly to `Core/AffineRecurrence.agda` via `Delta-bootstrap`, `U-bootstrap-closed`, and `tau-bootstrap-closed`; again the code's `Delta`-named arithmetic sequence is the paper's μ-sequence after the bridge identification of 3.10.
- 5.7 corresponds exactly to `Metatheory/Refactoring.agda` via `historical-support-correspondence` and `refactoring-invariance`.

No module cited in the theorem map above uses `postulate`. Postulates do occur in auxiliary modules elsewhere in the code base, but not in the modules invoked here.

This is a source-level mechanization claim, not a claim about public continuous integration coverage. The repository instructions give direct Agda type-check commands for the top-level artifact and tests, while the visible `PENEngineCI` workflow is an engine/Haskell workflow with Agda-harness contract checks rather than a direct `agdaPEN.agda` theorem-stack run. Thus public CI evidence should be read separately from the Agda kernel checking claim made here.

The trusted base is Agda with Cubical support together with the Cubical library declared in `pen.agda-lib`. The arithmetic layer uses `Core/Nat.agda`; the clutching formalization additionally uses the Cubical library's standard higher-inductive circle and suspension modules `Cubical.HITs.S1.Base` and `Cubical.HITs.Susp.Base`, together with the standard equivalence, isomorphism, and univalence infrastructure.

The formalization includes both direct cubical metatheory and a distilled extension calculus. The UIP collapse, horn-extension computation, the canonical-trace replacement interface, recent-history factorization chain, lower-bound obstruction, exact depth-two corollary, and clutching-family witness are mechanized against cubical or UIP primitives themselves. The remaining surface-syntactic theorems about normalized raw obligations, full tagged normalized signatures, and minimal opaque cost are then obtained in the paper by combining those formal metatheorems with the bridge normalization results of section 3, especially 4.9. The active-basis density, trace, refactoring, recurrence, and abstract 2D-foundations layers

package those results as the sealed-interface and counting calculus studied in the paper. The formalized core is therefore explicit, but the passage from raw declarations to the paper's horn and μ-based syntax still uses the bridge.

6.2. **Barrier discipline and affine recurrence.** Its organizing invariant is that resolved layers are exported through opaque interfaces. At the level of proof architecture, this is the exact content of 2.27, realized in Agda by `integration-trace-principle`: once an obligation has been discharged, later constructions depend only on the exported schema and not on the hidden derivation tree that discharged it. In the barrier modules, this is checked locally for the transitions around steps 8 and 9 by showing that inherited obligations are discharged from the owning layer's opaque record interface alone, without reopening deeper internal derivations.

This realizes the abstraction barrier promised by 2.10. In the formalization, 2.14 is exactly the package `transparent-growth-keeps-library-state`, `transparent-definitions-preserve-active`, `transparent-definitions-have-zero-integration-latency`, and `transparent-user-level-code-lies-o`. It is the reason historical depth is measured in terms of sealed layers rather than in terms of raw derivation ancestry.

On the counting side, the Agda development makes the modeling discipline explicit. The active primitive interface is represented as the disjoint union of the two most recent schema sets, and the saturation step is stated separately rather than hidden in the syntax of imports. In the basis-family presentation used here, recurrence counts the well-defined common cardinality of the normal-form transparent-generation classes for that interface and then reads that abstract count as minimal opaque cost via 3.10. The concrete decompositions in `Saturation/Decomposition.agda` then record, step by step, how the next obligation set splits into contributions from the two most recent sealed layers.

`Metatheory/UniversalRecurrence.agda` proves `universal-affine-recurrence`, while `Core/DepthOneAffine.agda` and `Core/AffineRecurrence.agda` package the exact arithmetic specializations `depth1-affine-growth`, `Delta-depth1-closed`, `tau-depth1-closed`, `Delta-bootstrap`, `U-bootstrap-closed`, and `tau-bootstrap-closed`. In particular, `Core/AffineRecurrence.agda` extends the payload-free counting law to the payload-aware recurrence

$$\mu_{n+1} = \mu_n + \mu_{n-1} + 2c,$$

defines the shifted sequence $U_n = \mu_n + 2c$, and proves that U satisfies the homogeneous Fibonacci recurrence. In the code base those arithmetic lemmas still carry `Delta`-names, because they are formalized on the abstract counting sequence before the bridge identification with μ. The zero-payload specialization is linked back to `Core/Nat.agda`, while `ObligationGraph/Recurrence.agda` records the payload-free corollary and `Test/Fibonacci.agda` checks the concrete Fibonacci identities.

The mechanization separates five claims that are easy to conflate in prose. The *abstraction-barrier claim* says that historical memory is mediated only by opaque exports. The *surface-image claim* says that raw structural obligations of the fixed extension calculus normalize into the horn-generated obligation language, rather than into an arbitrary closed filler space. The *metatheoretic depth claim* says that higher exact obligation types in that horn language are present but contractible at arity 3 and above, while binary obligations remain genuinely independent in general. The *syntactic minimality claim* says that, after passage through the bridge normalization theorems and the computational-replacement interface,

those present-but-contractible higher obligations are read as transparently derived fields and hence eliminated from μ-minimal public signatures, while the surviving binary ones satisfy recent-history factorization. The *recurrence claim* says that, once the depth-two window and explicit payload accounting are in place, the cardinality of the next primitive basis interface is governed by the affine depth-two law above. The formalization establishes the metatheoretic ingredients and the abstract counting law, and the paper records the bridge reading that turns them into the μ-based theorem stack.

Remark 6.1 (Scope of the mechanization)**.** The formalization establishes the cubical exact upper- and lower-bound metatheory, the abstract candidate-and-obligation calculus, the canonical-trace replacement interface, and the recurrence package built on them. The exact paper-to-Agda correspondences are listed above; the cited modules use no `postulate`; and the clutching module is checked separately only because its HIT imports live on the Cubical library's standard natural-number interface rather than on `Core/Nat.agda`. The mathematical arguments of section 4 remain important as explanation and as links to the categorical and topological examples. What remains at the paper level is the bridge from raw surface declarations to the canonical horn-obligation interface, the full tagged trace-presentation interface, and the μ-primitive filter, including 4.9, the surface-syntactic readings of 3.12, 4.13, 4.17, and the μ-based recurrence. This bridge is an image theorem for the fixed calculus, not a claim that the repository mechanizes a full parser for all of Cubical Agda.

7. Conclusion

The paper isolates *obligation depth* d_{obl} as the exact metatheoretic invariant of a fixed cubical extension calculus and distinguishes it from the separate minimal-signature depth d_μ governing what survives in μ-minimal normalized public form. The associated payload counts, abstract latency counts, and normalized-signature quantity μ are taken on counting normal forms of sealed interfaces, so they are stable under admissible record rebundling, interface slicing, and other canonical telescope refactorings. In extensional settings with UIP, higher coherences collapse and the obligation depth is 1. In the fixed fully coupled cubical extension calculus studied here, the obligation depth is 2, while 4.13 gives $d_\mu \leq 2$. Exact stabilization, elimination from minimal syntax, and chronological factorization are therefore three separate layers of the theorem architecture, not three phrasings of one fact. The main metatheorem is a depth-two affine recurrence for minimal opaque cost; under uniform payload size, the shifted sequence is Fibonacci.

The mechanism behind the recurrence is structural. In the formalization, 2.26 is read through `global-admissibility-forces-maximal-interface-density` as a contract-to-density theorem together with the bridge-level normalization theorems, 2.27 by `integration-trace-princi` and 4.18 by `two-layer-chronological-window`. Sealing cost is therefore identified with the size of the active basis interface only for extensions satisfying the fully coupled active-basis contract, while resolved coherence data disappear as derivations but remain present in the public signature as trace fields. In the cubical case, 4.9 first identifies marginal higher-arity raw structural data with based horn extensions rather than closed-sphere filler spaces; `structural-stabilizes-at-two` and `cubical-coherence-depth-exactly-two` make the exact stabilization statement precise, the replacement interface of `Metatheory/ComputationalReplacement.agda` turns cubically synthesized higher witnesses into elimination from the μ-minimal part of a canonical trace presentation, and `primitive-obligations-factor-through-last-two`

gives the recent-history factorization underlying the two-layer window. The explicit swap obstruction and the clutching family supply exact lower-bound witnesses. Under constant payload size `universal-affine-recurrence`, read through 3.10, specializes to the affine depth-two law and `Delta-bootstrap` gives the shifted Fibonacci corollary.

From this viewpoint the appearance of the Golden Ratio is not an empirical regularity of one implementation and not a matter of aesthetic numerology. It is the characteristic root of the homogeneous recurrence recovered after separating the constant payload contribution from the depth-two coherence memory.

The Cubical Agda formalization therefore certifies the cubical metatheory and abstract recurrence package on which the paper's extension-calculus theorem stack rests, rather than merely illustrating isolated ingredients. section 6 records the exact paper-to-Agda map; representative formal counterparts are `history-truncates-to-one` for the UIP collapse, `structural-stabilizes-at-two` and `cubical-coherence-depth-exactly-two` for the exact depth package, `higher-arity-computational-replacement` for the canonical-trace replacement step, `primitive-obligations-factor-through-last-two` for the recent-history factorization core, `cubical-depth-two-law-for-2d-foundations` for the abstract 2D-foundations wrapper, and `Delta-bootstrap`, read through 3.10, for the shifted Fibonacci corollary. The trust assumptions are exactly the Agda/Cubical kernel, the declared Cubical library, and the arithmetic layer used by the development, with no postulates in the cited proof modules; what remains outside the formal core is the paper-level image bridge from raw surface declarations to the canonical horn-obligation interface, the full tagged trace-presentation interface, and the μ-primitive filter. A natural next step is to connect this metatheory more tightly to concrete systems for generating and integrating candidate extensions.

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